Stress-testing SURD for AGN Reverberation Mapping: NGC 5548

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We stress-test multivariate information decomposition for reverberation mapping using the 5100-Å continuum and velocity-resolved Hβ light curves of NGC 5548. On a strict-overlap window (JD less 2400000: 47512–49255), a bidirectional interpolated cross-correlation analysis with flux-randomization/random-subset-sampling uncertainties gives centroid delays of $19.44^{+1.52}_{-1.49}$ d (blue wing), $22.79^{+1.85}_{-1.80}$ d (core), and $30.30^{+4.52}_{-3.62}$ d (red wing). By contrast, three-predictor Synergistic–Unique–Redundant Decomposition (SURD) scans have range-dependent maxima at 93–198 d. A matched 999-realization circular-shift test finds none significant ($p_{\rm global}$ = 0.873–1.000), and empirical lagwise p-values yield no Benjamini–Hochberg discoveries. Matched synthetic ablations show that histogram dimensionality, red-noise structure, the observing window, interpolation, and normalization are jointly sufficient to produce false long-lag maxima. A target-history-conditioned maximum at 73 d is also insignificant ($p_{\rm global}$ = 0.9802); changing between 3–8 equal-width or quantile bins moves it to 37–102 d while leaving every global test non-significant. Continuous nearest-neighbour estimates recover significant prompt multivariate dependence for all three line targets, but no significant 61–200 d or history-conditioned maximum. As an external check, native-cadence HST/COS light curves of Mrk 817 show elevated information near published ultraviolet-line delays, but unrestricted maxima are generally unstable across estimator scale and time-pairing tolerance. We conclude that the long-lag SURD features are not astrophysically established. Information-decomposition scans of irregular AGN data require matched controls, estimator cross-checks, and scan-wide significance tests before individual atoms or lags receive physical interpretation.

Key words: galaxies: active – galaxies: nuclei – galaxies: individual: NGC 5548 – galaxies: individual: Mrk 817 – methods: statistical

1 INTRODUCTION

Active galactic nuclei (AGN) are powered by accretion onto supermassive black holes and radiate over a broad range of wavelengths through a compact but highly structured central engine. The standard observational picture includes an accretion disk that produces much of the optical and ultraviolet continuum, a hot corona responsible for the X-ray emission, a broad-line region (BLR) composed of dense photoionized gas moving at velocities of order 10^3 to 10^4 km s^{-1}, and more distant dusty material that reprocesses the radiation field at infrared wavelengths (Peterson 2004; Netzer 2013; Cackett et al. 2021). Because these regions are too compact to resolve directly in most AGN, temporal variability provides one of the few effective ways to infer their sizes, structure, and coupling.

Reverberation mapping exploits the fact that fluctuations in the central ionizing source are echoed by delayed responses in remote reprocessing or emission-line regions. In its simplest linearized form, the response of an emission line $L(t)$ to a driving continuum $C(t)$ may be written as

$$L(t) = \int \Psi(\tau)\, C(t-\tau)\, d\tau, \tag{1}$$

where $\Psi(\tau)$ is the transfer function or delay distribution. The classical use of this formalism has been to derive a characteristic lag τ, associate it with a BLR radius $R \sim c\tau$, and combine that scale with the line width to estimate the black-hole mass under a virial assumption (Peterson 2004; Bentz & Katz 2015). This framework has been extremely successful and remains foundational to AGN black-hole mass scaling relations.

Nevertheless, the astrophysical interpretation of reverberation data has become progressively more subtle. The quantity that drives the broad emission lines is the ionizing continuum, particularly in the extreme ultraviolet and soft X-ray bands, whereas the continuum actually monitored in many optical reverberation campaigns is a more accessible proxy, often the flux near 5100 Å. As a result, the measured lag between an observed continuum band and an emission line need not correspond to a direct causal connection between those two observables alone. Instead, both may be responding, with different delays and different filtering, to a latent radiation field that is only imperfectly traced by the observed bands (Cackett et al. 2021). The question is therefore not merely what the lag is, but whether the monitored continuum contains the relevant information needed to explain the line response.

This issue is underscored by continuum reverberation mapping. A series of high-cadence monitoring programs has revealed that ultraviolet and optical continuum bands in AGN are delayed with respect to one another, with delays that often grow systematically with wavelength as expected for a thermally stratified accretion disk. However,

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the measured lags are frequently larger than predicted by the simplest geometrically thin, optically thick accretion-disk models (Fausnaugh et al. 2016; Fausnaugh 2017; Edelson et al. 2019; Cackett et al. 2021). These results suggest that the observed continuum variability is not fully described by a minimal reprocessing picture and may involve additional processes such as intrinsic accretion-rate fluctuations, diffuse continuum emission, complex irradiation geometry, or nontrivial radiative transfer. From the standpoint of BLR reverberation, this implies that optical continuum light curves may not be interpreted as pristine drivers without further caution.

NGC 5548 is the canonical source in which these concerns become impossible to ignore. It is one of the most intensively monitored Seyfert galaxies and has served as a cornerstone of reverberation mapping for decades. It has played a major role in calibrating broad-line lags, virial mass estimates, and radius–luminosity relations, while more recent campaigns have transformed it into a benchmark for multiwavelength and velocity-resolved reverberation analysis (Peterson et al. 2002; Bentz & Katz 2015; Cackett et al. 2021). In particular, the AGN STORM campaign delivered one of the richest reverberation datasets ever obtained, including optical and ultraviolet continuum monitoring, extensive spectroscopy, and detailed velocity-delay information for several broad lines (De Rosa et al. 2015; Pei et al. 2017; Horne et al. 2021).

These observations revealed that NGC 5548 is not well described by a stationary one-zone BLR driven by a single observed continuum proxy. Velocity-resolved reverberation results showed clear structure across the Hβ profile, with the line wings and core participating differently in the reverberation pattern (Pei et al. 2017; Horne et al. 2021). The interpretation of these data points toward a BLR that is radially stratified and kinematically complex, possibly including multiple components or zones rather than a single homogeneous distribution of clouds (Horne et al. 2021; Long & Dexter 2025). This immediately motivates a shift away from purely integrated-line analyses. If the blue wing, line core, and red wing carry different dynamical information, then treating Hβ as a single scalar time series necessarily compresses part of the astrophysical content of the data.

An even more striking complication emerged during the same source's monitoring history: the so-called "holiday" in NGC 5548. During part of the AGN STORM campaign, the broad emission lines became partially decorrelated from the observed UV continuum, even though they had previously tracked it closely. The leading interpretation is that an obscuring wind or shielding structure modified the ionizing spectral energy distribution reaching the BLR while leaving the observed UV continuum comparatively unaffected (Dehghanian et al. 2019). The importance of this result cannot be overstated. It demonstrates directly that an observed continuum channel may cease to be a faithful tracer of the radiation field relevant for BLR photoionization. Consequently, a line–continuum lag analysis may look physically incomplete not because the line has stopped reverberating, but because the monitored continuum is no longer the appropriate driver proxy.

This broader lesson is now reinforced by work across the reverberation-mapping literature. Velocity-resolved Hβ studies increasingly show that BLR kinematics are diverse across AGN and may evolve over time, with some sources displaying signatures consistent with inflow, others with outflow, and others with virialized or mixed dynamics (Du et al. 2018; Wang et al. 2025; Feng et al. 2024). In parallel, changing-look AGN have demonstrated that the relation between continuum state and broad-line emission can undergo dramatic reorganization on humanly accessible timescales (Ricci & Trakhtenbrot 2023). Although changing-look phenomena are not the focus of the present study, they help establish a wider astrophysical principle: the mapping between observed continuum, ionizing continuum, and line-emitting gas is not guaranteed to be stationary.

Within this context, NGC 5548 is particularly well suited for a conservative test of multivariate reverberation analysis. The source offers high-quality continuum monitoring, extensive Hβ spectroscopy, and a rich legacy of prior lag measurements against which information-based diagnostics can be checked. Treating the blue wing, core, and red wing as distinct observables permits questions about joint predictability that an integrated line curve cannot address. It also creates substantial statistical risk: autocorrelation, seasonal gaps, interpolation, multiple lag searches, and high-dimensional density estimation can all generate apparent structure. NGC 5548 therefore provides both an astrophysical application and a demanding stress test of the method.

These questions are not purely methodological. They arise naturally from long-standing astrophysical uncertainties about the BLR and the continuum source. The BLR is expected to be sensitive to the ionizing continuum, but in practice the ionizing continuum is usually unobserved. The optical continuum is measurable but is itself part of a reverberating accretion-flow system and may therefore contain both intrinsic and reprocessed variability components (Fausnaugh 2017; Cackett et al. 2021). The Hβ line is observable with high fidelity, but its integrated flux merges together multiple kinematic regions whose relative responses may differ. Once velocity-resolved spectroscopy is available, the astrophysical challenge becomes one of causal disentanglement among observables that are neither independent nor trivially ordered in a simple driver–response chain.

A number of established approaches address parts of this challenge. Interpolated cross-correlation methods and related lag estimators remain the standard tools for continuum–line reverberation analysis and continue to provide essential baseline measurements (Peterson 2004; White & Peterson 1994). JAVELIN and related stochastic-process approaches model reverberation as a damped random walk continuum filtered by a transfer function, thereby improving lag estimation and uncertainty quantification in many cases (Zu et al. 2011, 2013). Velocity-delay mapping and dynamical modeling aim to reconstruct the geometry and kinematics of the BLR directly from the line-profile response (Pancoast et al. 2014; Horne et al. 2021). Continuum reverberation studies focus on interband disk lags and the thermal structure of the accretion flow (Fausnaugh et al. 2016; Fausnaugh 2017; Edelson et al. 2019). Each of these approaches has been productive, but they are typically tailored to a specific projection of the problem: pairwise lags, transfer-function fitting, dynamical inversion, or continuum size scaling.

What remains comparatively underexplored is the specific astrophysical question of how multiple simultaneously observed channels jointly constrain the future evolution of one chosen component of the system, especially when those channels include both the continuum and velocity-resolved line segments. In other words, once the Hβ profile has been decomposed into blue wing, core, and red wing light curves, and once a conservative common time interval is imposed, the natural problem is no longer simply "what is the lag?" It becomes: how much of the future behavior of one velocity component is already encoded in the continuum, how much is encoded in another part of the line profile, and how much lies beyond the information available in the monitored observables? That is the astrophysical problem motivating the present paper.

The emphasis on strict temporal overlap is also not incidental. Irregular cadence, seasonal gaps, and unequal overlap among observed channels are intrinsic features of AGN monitoring data. Any multivariate time-series analysis is therefore vulnerable to spurious structure introduced by aggressive interpolation, extrapolation across

gaps, or implicit assumptions about the simultaneity of channels that were not truly observed together. In the context of NGC 5548, where the scientific questions depend sensitively on comparing the optical continuum to distinct Hβ velocity components, it is essential to work under conservative overlap conditions. The goal is not to maximize sample size at all costs, but to preserve physical interpretability. A shorter but genuinely shared time interval is preferable to a longer one in which apparent multivariate structure may be partly synthetic.

The aim of this paper is to determine which conclusions from a multivariate SURD analysis remain defensible after conservative temporal overlap, synthetic recovery tests, red-noise-preserving global nulls, discretization sensitivity, and a continuous-estimator cross-check. We retain the blue wing, line core, and red wing as separate channels, but treat any inferred atom or lag as exploratory unless it is stable under those controls.

The methodological framework used later in the paper is based on information decomposition, but we intentionally postpone its formal description. The present Introduction has a different role: to show why such an analysis is astrophysically warranted. The literature has already established that NGC 5548 exhibits continuum interband lags, velocity-resolved Hβ structure, partial line–continuum decoupling during the holiday, and evidence for a multicomponent BLR. The natural next step is not simply to add another lag estimator, but to investigate the reverberation data in a way that is explicitly multivariate, velocity-resolved, and conservative with respect to temporal overlap. That is the problem we address here.

The structure of this paper is as follows. Section 2 introduces the information framework; Section 3 describes the observations and strict-overlap processing; Section 4 presents the tests and results; Section 5 discusses their implications; and Section 6 summarizes the conclusions.

2 SYNERGISTIC–UNIQUE–REDUNDANT DECOMPOSITION (SURD) FOR TIME SERIES

2.1 Problem formulation

Consider a collection of N observed stochastic processes,

$$\mathbf{Q}(t) = \big(Q_1(t), Q_2(t), \ldots, Q_N(t)\big), \tag{2}$$

sampled at discrete times $t = t_n$. For a chosen target index j, the goal is to quantify how much information about the future state

$$Q_j^+ \equiv Q_j(t + \Delta T) \tag{3}$$

is contained in the observed variables at time t and, crucially, to decompose that predictive information into *redundant*, *unique*, and *synergistic* contributions (Martínez-Sánchez et al. 2024).

The central motivation is that in multivariate systems, pairwise predictive measures such as Granger causality or transfer entropy can establish that a variable is informative about a target, but they do not in general distinguish whether that information is:

(i) unique to one predictor,
(ii) duplicated across several predictors, or
(iii) available only from a joint observation of several predictors.

SURD is designed precisely to perform that decomposition in a time-series setting (Martínez-Sánchez et al. 2024; Schreiber 2000; Williams & Beer 2010).

2.2 Information-theoretic preliminaries

Let $Y \equiv Q_j^+$ denote the target and let $\mathbf{X}$ denote the vector of observed predictors. For discrete variables, the Shannon entropy is

$$H(Y) = -\sum_y p(y) \log_2 p(y), \tag{4}$$

and the conditional entropy of Y given $\mathbf{X}$ is

$$H(Y \mid \mathbf{X}) = -\sum_{y,\mathbf{x}} p(y, \mathbf{x}) \log_2 p(y \mid \mathbf{x}). \tag{5}$$

The mutual information between Y and $\mathbf{X}$ is

$$I(Y; \mathbf{X}) = H(Y) - H(Y \mid \mathbf{X}) = \sum_{y,\mathbf{x}} p(y, \mathbf{x}) \log_2 \frac{p(y, \mathbf{x})}{p(y)p(\mathbf{x})}. \tag{6}$$

This quantity measures the total predictive information about the future target that is available in the observed present state (Shannon 1948; Cover & Thomas 2006).

In SURD, the unexplained part of the future is interpreted as an *information leak*,

$$\Delta I_{\text{leak}\to j} = H(Y \mid \mathbf{X}), \tag{7}$$

so that the forward information balance reads

$$H(Y) = I(Y; \mathbf{X}) + \Delta I_{\text{leak}\to j}. \tag{8}$$

Equation (8) is one of the core ideas of SURD: the future uncertainty of the target splits into an observed part, explained by the measured variables, and an unobserved part, attributed to hidden or unmeasured influences (Martínez-Sánchez et al. 2024).

2.3 From total predictive information to information-decomposition components

Let the predictor vector be

$$\mathbf{X} = (X_1, \ldots, X_M), \tag{9}$$

where the X_α may correspond either to different physical variables at the same time, or to different lagged copies of the same variables. SURD decomposes the observable predictive information $I(Y; \mathbf{X})$ into additive components:

$$I(Y; \mathbf{X}) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j}, \tag{10}$$

where:

- $\Delta I^R_{\mathcal{A}\to j}$ is the *redundant* contribution from the group of predictors indexed by $\mathcal{A}$,
- $\Delta I^U_{\alpha\to j}$ is the *unique* contribution of predictor X_α,
- $\Delta I^S_{\mathcal{A}\to j}$ is the *synergistic* contribution from the group $\mathcal{A}$,
- C denotes all non-singleton subsets of $\{1, \ldots, M\}$.

The full future entropy therefore obeys

$$H(Y) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j} + \Delta I_{\text{leak}\to j}. \tag{11}$$

The three information-decomposition types have the following interpretation (Martínez-Sánchez et al. 2024):

- Redundant: the same predictive information is shared across several predictors,

- Unique: a predictor contains information not available from any other predictor alone,

- Synergistic: information appears only when several predictors are observed jointly.

For the three-predictor scans below, "total synergy" denotes $S_{12}+S_{13}+S_{23}+S_{123}$ and "total redundancy" denotes $R_{12}+R_{13}+R_{23}+R_{123}$. These sums, rather than any single atom, are the quantities shown in the primary lag scans.

The statewise construction and detailed lagged formalism follow Martínez-Sánchez et al. (2024); we omit that derivation here and retain only the quantities required for the analysis.

2.4 Normalization and interpretation

Because the observable predictive information is conserved by construction, the unique, redundant, and synergistic terms are naturally normalized by the total observable mutual information:

$$\widehat{\Delta I}^{R}_{\mathcal{A}\to j} = \frac{\Delta I^{R}_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^{U}_{\alpha\to j} = \frac{\Delta I^{U}_{\alpha\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^{S}_{\mathcal{A}\to j} = \frac{\Delta I^{S}_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}. \tag{12}$$

These normalized terms sum to unity:

$$\sum_{\mathcal{A}\in C} \widehat{\Delta I}^{R}_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \widehat{\Delta I}^{U}_{\alpha\to j} + \sum_{\mathcal{A}\in C} \widehat{\Delta I}^{S}_{\mathcal{A}\to j} = 1. \tag{13}$$

The leak is normalized instead by the total entropy of the target:

$$\widehat{\Delta I}_{\mathrm{leak}\to j} = \frac{\Delta I_{\mathrm{leak}\to j}}{H(Y)} = \frac{H(Y \mid \mathbf{X})}{H(Y)}. \tag{14}$$

Hence:

- $\widehat{\Delta I}_{\mathrm{leak}\to j} \approx 0$ means the observed predictors account for almost all predictive information about the target,
- $\widehat{\Delta I}_{\mathrm{leak}\to j} \approx 1$ means most of the target's future uncertainty remains unexplained by the observed variables.

2.5 Practical estimation

In applications, the probabilities entering Eq. (6) and the statewise SURD construction must be estimated from finite data. The original SURD work uses histogram-based and transport-map-based estimators depending on dimensionality and sample size (Martínez-Sánchez et al. 2024). In a histogram implementation, one discretizes the predictor-target space into bins and estimates

$$p(y,\mathbf{x}) \approx \frac{n(y,\mathbf{x})}{N_{\mathrm{samples}}}, \tag{15}$$

where $n(y,\mathbf{x})$ is the bin count. The resulting empirical joint distribution is then used to compute all required marginal and conditional probabilities. More sophisticated estimators, such as transport maps, are advantageous in high dimensions because the number of bins required by direct histograms grows exponentially with the number of predictors.

Algorithmically, the SURD workflow for a target Q_j^+ is:

(i) choose the predictor vector $\mathbf{X}$, including variables, lags, or contemporaneous channels;
(ii) estimate the probability distributions $p(y,\mathbf{x}_{\mathcal{A}})$ for all relevant predictor subsets $\mathcal{A}$;
(iii) compute the specific informations $\tilde{\imath}_{\mathcal{A}}(y)$ for every target state y;
(iv) decompose the statewise increments into redundant, unique, and synergistic parts;
(v) average over y to obtain ΔI^R, ΔI^U, and ΔI^S;
(vi) compute the leak from Eq. (7);
(vii) normalize the observable components by $I(Y;\mathbf{X})$ and the leak by $H(Y)$.

3 ADOPTED DATA, SPECTRAL PROCESSING, AND SAMPLING VALIDATION

The adopted reduction uses 242 Hβ spectra on 224 distinct dates. Of the 247 archived profiles, five lack an exact date and instrument match in corrected Table 1 of Wanders & Peterson (1996) and are excluded under a documented provenance policy. These are n57653ab.spc, n57654ab.spc, n57657ab.spc, n57711ab.spc, and n57725ab.spc. Their original files are preserved; the exclusion does not establish that their measurements are invalid.

We express dates as JD less 2400000, rather than MJD. The revised continuum file requires 40000 to be added to its tabulated dates, despite the archive landing page stating the other convention. All 242 selected dates and revised continuum fluxes have unique matches in the corrected table within the documented flux tolerance of 0.00051 in the revised archive units. Spectral dates have integer day precision; the precise time scale and correction to the heliocentre remain unspecified.

The adopted velocity intervals are $[-6000,-2000)$, $[-2000,2000)$ and $[2000,6000)$ km s^{-1} for the blue wing, core and red wing respectively. We use observed wavelengths with reference wavelength 4861.33 Å and numerical reference redshift 0.017175. These conventions retain the existing project reduction; they are not the exact intervals or redshift used in the source paper. Each selected pixel contributes its flux density multiplied by its 2 Å width. Profiles are in arbitrary flux density units, so the resulting components are in arbitrary integrated units.

Measurements sharing a date are averaged with equal weights for all components, preserving the identity between their sum and the integrated selected profile. Errors propagate in quadrature under independence assumptions. Pixel covariance and shared calibration covariance are unavailable. The selected profile sum differs from the archive total Hβ light curve, whose subtraction conventions differ.

The original continuum remains primary for continuity and denser observed sampling. The revised continuum is retained as a sensitivity input. Its flux units and galaxy subtraction differ from the original continuum. Table 1 separates measurement counts from distinct dates. Each series retains its observed endpoints; lagged pairing enforces the support of each required series.

We compare a daily linear interpolation grid, interpolation restricted to gaps of at most 30 days, analysis within seasons divided at gaps exceeding 60 days in either input and also restricted to 30 day interpolation gaps, and native observation pairing with a one day tolerance. Native ties select the earlier observation. No method extrapolates. A positive lag places the line target after its predictors. A native target contributes at most one tuple at each lag, although predictor observations may be reused. Counts therefore do not represent independent sample sizes (Table 2).

Series	Measurements	Dates	First JD offset	Last JD offset
Hβ components	242	224	47512.00	49255.00
Original continuum	557	557	47512.00	49243.66
Revised continuum	242	224	47512.00	49255.00

Table 1. Adopted input counts and actual support endpoints. JD offsets are relative to 2400000. The revised continuum is a sensitivity input.

Lag (days)	Daily	Limited gaps	Within seasons	Native
0	1732	1363	1363	223
30	1714	1174	1174	74
100	1644	980	980	60
200	1544	840	778	51

Table 2. Lagged tuple counts under the adopted sampling rules. There are 224 native line dates and 557 original continuum dates.

The exploratory information pilot scans 201 lags from zero through 200 days for three line targets. Each target uses continuum and the other two line components at a common lag as predictors; the target component is excluded at every lag. Four marginal quantile bins are defined from native observations and held fixed across methods and lags within each dataset. The identical rule defines bins for each simulated dataset. A minimum of 32 tuples is a numerical guard, not evidence of adequate statistical support. Joint mutual information, individual SURD atoms, total synergy in bits, synergy divided by joint information, and histogram occupancy are saved separately.

Eight simulated datasets have exercised this complete execution path, using independent signals and a delayed shared continuum driver. The process parameters are stipulated and have not been fitted to the observed campaign. This is a runtime and correctness pilot, not significance calibration or a detection power measurement. These pilot scans are separate from the results of the next section, which are computed on the adopted sample.

4 INFORMATION DECOMPOSITION AND LAG ANALYSIS RESULTS

We run the SURD algorithm in a symmetric, round-robin configuration on the prepared NGC 5548 data to study predictive dependence between the four observables: the continuum, the red and blue wings, and the core of Hβ. For each run, we set one observable as the target Y and use the remaining three observables as the predictor set $\mathbf{X} = \{X_1, X_2, X_3\}$. The continuum-target run is included only as a reverse-prediction diagnostic of the estimator; it is not a physical reverberation configuration in which the line drives the continuum.

4.1 Dominant Lags and Extended Scan Range

To test the physical stability of the synergy and leak profiles, we extended the lag search range from the initial limit of 60 days to 200 days. The comparison of peak lags and synergy values between the standard and extended scans is summarized in Table 3, and the complete information decomposition profiles are displayed in Figure 2.

For all four observables, the peak synergy lag shifts when the scan range is extended from 60 days to 200 days. Specifically, the Continuum peak shifts from 34 days to 115 days, the Red Wing peak shifts from 46 days to 198 days, the Core peak shifts from 53 days to 128 days, and the Blue Wing peak shifts from 56 days to 93 days. This variability shows that the unconditioned peaks are scan-range-dependent and do not converge on stable physical delays; boundary and windowing effects are therefore plausible alternatives to a physical interpretation.

The middle column supplies an important qualification to the normalization argument. The estimated joint mutual information does not approach zero: across targets and lags it spans 1.14–1.97 bits, and at the four normalized maxima it is 1.25–1.66 bits. The corresponding absolute total synergy is 0.90–1.17 bits. Normalization changes the relative shape, especially where joint MI declines, but division by an almost-zero denominator is not an adequate explanation of the real curves. These large absolute values under sparse four-dimensional histograms are themselves estimator-biased, as shown by the matched independent-red-noise control below. We therefore base inference on matched empirical nulls rather than on the height of either curve.

4.2 Surrogate and Null Hypothesis Testing

To test the exact primary statistic, we generate 999 null realizations in which all four interpolated series are independently circularly shifted by at least 60 d. Each realization repeats the same three-predictor dimensionality, eight-bin equal-width histogram, strict-overlap sample window, normalization, 1–200 d scan, and maximum statistic used for Figure 2. Circular shifts destroy relative phase alignment while preserving each series' values and cyclic autocorrelation. Figure 3 shows the resulting matched pointwise envelope for the core target and a matched three-predictor bin-count sensitivity test.

Crucially, after applying the Benjamini–Hochberg False Discovery Rate (FDR) correction to the main 3-predictor lag scans over 200 tested lags, **none of the individual lag bins remains globally significant.** This is the key methodological result: local exceedance above pointwise surrogate envelopes is not sufficient grounds for claiming a physically meaningful delay. Instead, the observed SURD curves should be interpreted as broad lag-dependent structures whose most prominent maxima remain compatible with scanned red-noise nulls once multiple testing is taken seriously.

4.3 Comparison with Classical Reverberation Lags

To compare our information-theoretic results with a standard reverberation estimator, we compute the ICCF directly from the native unevenly sampled continuum and line light curves. At each trial lag from −100 to +200 d in 1-d steps, we linearly interpolate each light

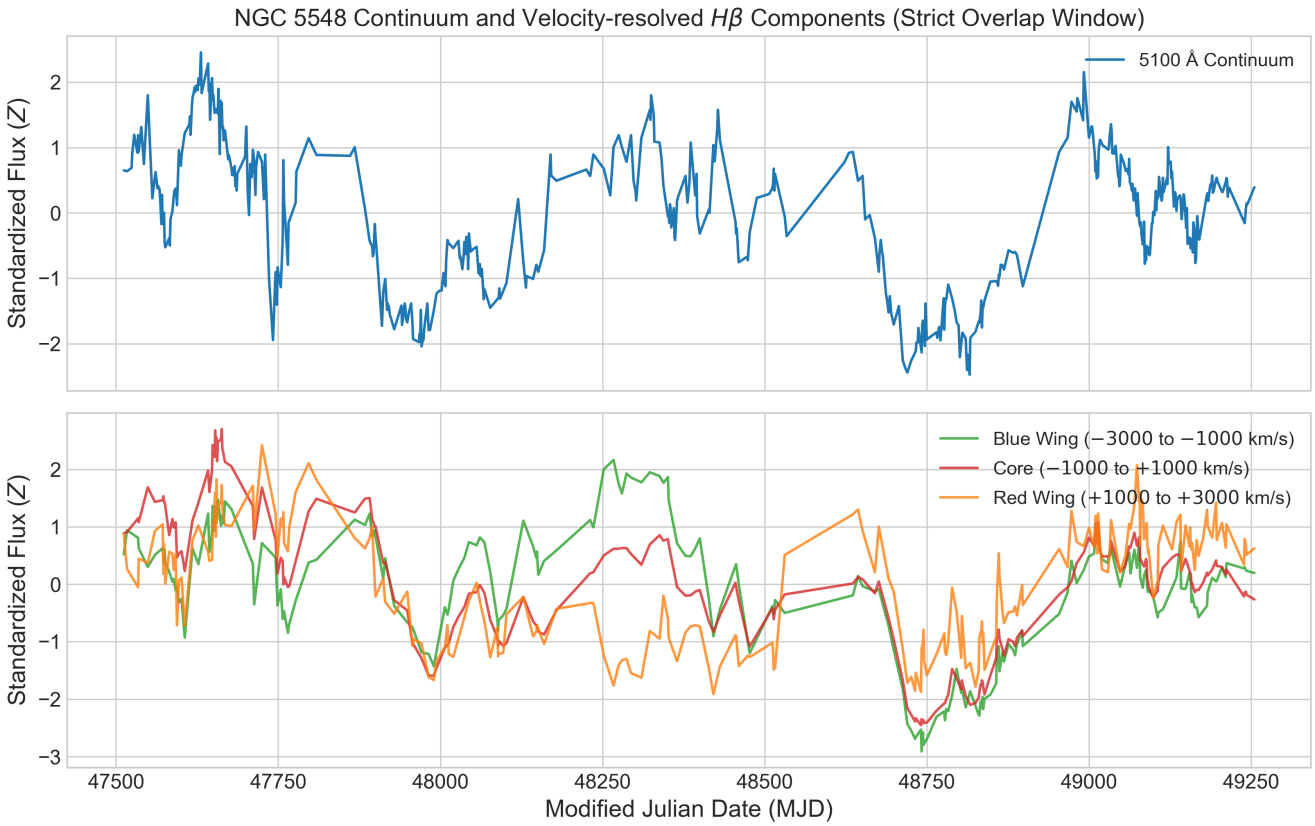


Figure 1. Top panel: Standardized 5100 Å continuum light curve for NGC 5548 over the strictly aligned overlap window (JD less 2400000: 47512–49255). Bottom panel: Standardized velocity-resolved Hβ components (Blue Wing, Line Core, and Red Wing) resampled onto the shared 1.0-day grid.

Target Component	Old Peak Lag (d)	New Peak Lag (d)	Old Peak Synergy	New Peak Synergy
Continuum	34	115	0.6882	0.7138
Red Wing Hβ	46	198	0.7218	0.7358
Core Hβ	53	128	0.6330	0.7045
Blue Wing Hβ	56	93	0.6787	0.7133

Table 3. Comparison of peak normalized synergy lags and values between the standard (1–60 days) and extended (1–200 days) round-robin 3-predictor SURD scans. Note that the peaks shift substantially when extending the scan window, indicating that the unconditioned peak lag is scan-range-dependent and does not represent a stable physical delay.

curve to the shifted epochs of the other and average the two correlation coefficients. We define the centroid from the contiguous region around the primary peak with $r \geq 0.8 r_{\max}$ and estimate uncertainties from 5000 flux-randomization/random-subset-sampling (FR/RSS) realizations (White & Peterson 1994; Peterson 2004). Positive lag means that the line follows the continuum.

The blue-wing and core centroid distributions are compact and prompt. The red-wing ICCF has a native peak at 15 d but a broad above-threshold shoulder, producing a substantially later centroid of $30.30^{+4.52}_{-3.62}$ d. We therefore do not infer velocity-resolved inflow or outflow from these three summaries. In contrast, the normalized three-predictor SURD maxima occur at 93–198 d and fail the matched global null tests. The leak minima are also descriptive diagnostics rather than lag estimates.

Figure 4 makes the main contrast visually explicit: pairwise continuum–line correlation peaks remain prompt, whereas the normalized multivariate synergy curve drifts to much longer lags and shows the scan-range dependence documented above. This separation is one of the clearest indications that the long-lag SURD maxima are not tracing the same physical structure as the classical reverberation lag.

4.4 Empirical Multiple-Testing Correction

For each target and lag, we use the empirical one-sided p-value

$$p(\tau) = \frac{1 + \#\{\widehat{S}_b(\tau) \geq \widehat{S}_{\mathrm{obs}}(\tau)\}}{1 + B}, \tag{16}$$

where $B = 999$ is the number of matched circular-shift realizations. No distributional approximation is used. Benjamini–Hochberg adjustment across the 200 lags for each target yields no discoveries at $q = 0.05$; the smallest unadjusted empirical p-values are 0.694,

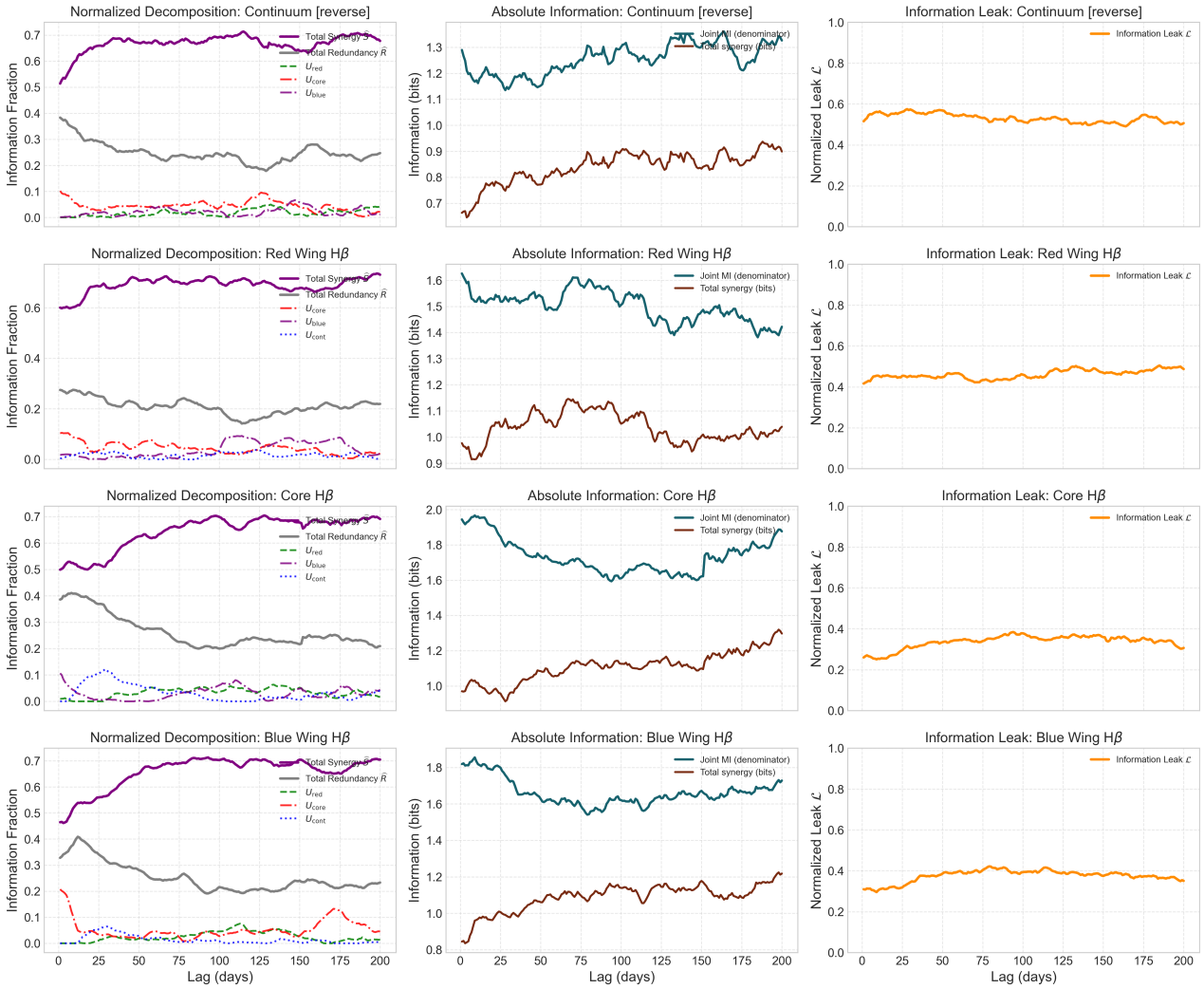


Figure 2. Three-predictor round-robin scans. The left column shows normalized atoms, which sum to unity. The middle column shows total joint mutual information, the normalization denominator, together with absolute total synergy in bits. The right column shows normalized information leak $\mathcal{L} = H(Y \mid X_1, X_2, X_3)/H(Y)$. Total synergy is $S_{12} + S_{13} + S_{23} + S_{123}$, with an analogous sum for redundancy. The continuum-target row is a reverse-prediction estimator diagnostic, not a physical line-to-continuum reverberation model.

Target Component	ICCF centroid (d)	ICCF peak (d)	SURD max (d)	Leak min (d)
Blue Wing Hβ	$19.44^{+1.52}_{-1.49}$	19	93	9
Core Hβ	$22.79^{+1.85}_{-1.80}$	22	128	9
Red Wing Hβ	$30.30^{+4.52}_{-3.62}$	15	198	1

Table 4. Native-cadence bidirectional ICCF centroid lags with 16th–84th percentile FR/RSS uncertainties, native ICCF peak lags, and descriptive locations of the normalized three-predictor SURD maximum and leak minimum. The SURD locations are not significant lag measurements.

0.260, 0.354, and 0.349 for the continuum, red, core, and blue targets, respectively.

To test the selected maximum directly, each of the same 999 realizations contributes its maximum over the full 200-lag scan. The empirical global p-value is

$$p_{\mathrm{global}} = \frac{1 + \#(\widehat{S}_{\mathrm{max,null}} \geq \widehat{S}_{\mathrm{max,real}})}{1 + N_{\mathrm{surrogate}}}. \tag{17}$$

The observed maxima are 0.7138, 0.7358, 0.7045, and 0.7133 for the continuum, red, core, and blue targets. Their global p-values are 1.000, 0.873, 0.947, and 0.936, respectively. The observed maxima are therefore consistent with the scanned red-noise null.

Thus, although local features may appear prominent, none survives the matched scan-wide controls. Empirical multiple-testing corrections and negative controls are necessary before isolated information-curve maxima are interpreted as lags.

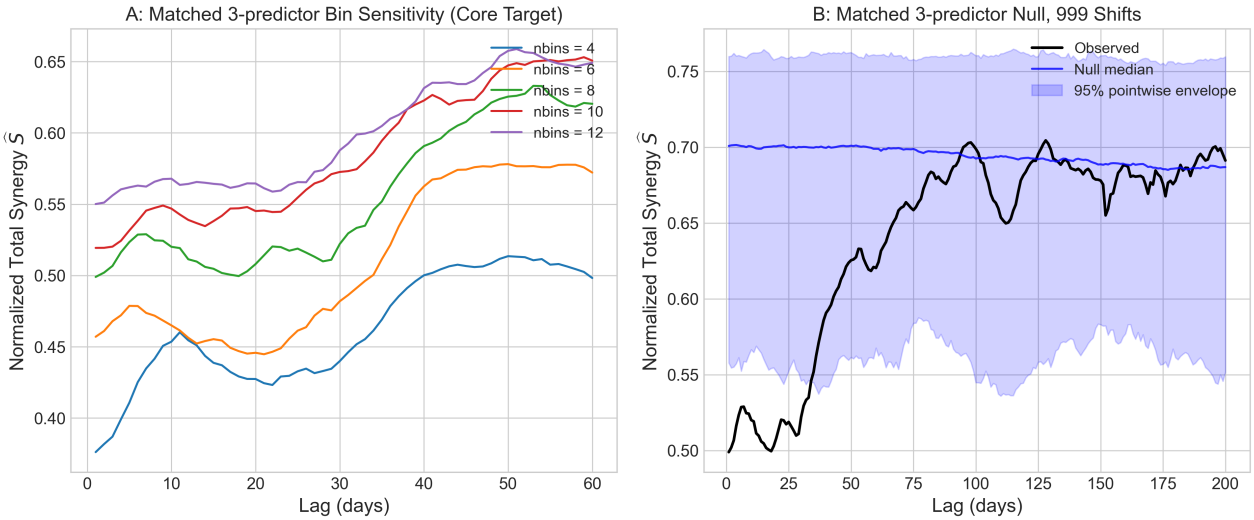


Figure 3. Matched three-predictor robustness checks for the core target. Panel A varies the equal-width histogram from 4 to 12 bins over 1–60 d. Panel B compares the observed normalized total synergy with the median and pointwise 95% envelope from 999 circular-shift realizations. The global test uses each realization's maximum over the full 1–200 d scan rather than this pointwise envelope.

4.5 Continuum-Only Baseline and Prompt Lag Recovery

The three-predictor round-robin SURD scans in Section 4 exhibit unconditioned maxima at long lags (93–198 d) that fail empirical global significance tests against circular-shift surrogates (p_{global} = 0.873–1.000). A natural question arises: can information-theoretic estimators recover the known physical reverberation delay (∼ 15–25 d) when the line emission is evaluated directly against the optical continuum driver, without the confounding influence of correlated line predictors?

To test this, we evaluate the one-predictor mutual information baseline:

$$I(Y(t); F_{5100}(t-\tau)) \tag{18}$$

across lags $\tau \in [0, 200]$ d for the three Hβ components (Blue Wing, Core, and Red Wing). We evaluate this baseline across all four adopted sampling configurations: daily interpolation, interpolation restricted to gaps ≤ 30 d, within-season interpolation, and native observation pairing.

Under gap-limited sampling (which bridges gaps ≤ 30 d without interpolating across seasonal observing shutdowns), the continuum-only scan identifies prompt mutual information maxima for all three components (Table 5):

- **Blue Wing:** τ_{peak} = 16 d (I = 0.473 bits, leak minimum $\mathcal{L}$ = 0.762 at 16 d).
- **Line Core:** τ_{peak} = 15 d (I = 0.719 bits, leak minimum $\mathcal{L}$ = 0.637 at 15 d).
- **Red Wing:** τ_{peak} = 23 d (I = 0.326 bits, leak minimum $\mathcal{L}$ = 0.834 at 23 d).

Under within-season sampling, the peak lags and mutual information values are identical to the gap-limited results. Under daily interpolation, the recovered peaks are 20 d (Blue), 16 d (Core), and 13 d (Red). Under native observation pairing, peaks occur at 26 d (Blue), 25 d (Core), and 67 d (Red), although native counts are limited to 61–68 tuples per lag.

We calibrate significance using N = 999 circular-shift null realizations and the scan-wide maximum statistic. Blue has p_{global} = 0.009 and q = 0.0135, while Core has p_{global} = 0.001 and q = 0.003. Red lies within the null distribution ($p_{\text{global}} = q$ = 0.228). These statements apply to the selected gap-limited estimator and null; they do not correct across every exploratory sampling and estimator choice in the repository.

In synthetic simulations with an injected 20-day delay, the one-predictor continuum baseline cleanly recovers the true delay (17–22 d) across all four random seeds and all sampling methods. This establishes a key methodological insight: mutual information is fully capable of resolving prompt reverberation delays, but the unconditioned three-predictor SURD setup suffers from inter-line cross-correlations that create spurious long-lag maxima when line components are used as predictors of one another.

4.6 Calibration of False Positives and Detection Power

A rigorous application of information theory to observational astronomical data requires determining the detection threshold and false positive rate under realistic campaign conditions. To quantify the sensitivity limits of the 1988–1993 NGC 5548 observing window, we conduct a controlled simulation grid across response amplitudes and delays.

We simulate synthetic Damped Random Walk (DRW) continuum drivers (τ_{DRW} = 50.0 d) and generate delayed line responses:

$$y(t) = A \cdot \bar{x}_{\text{delayed}}(t) + (1 - A) \cdot \eta_{\text{indep}}(t), \tag{19}$$

where $A \in [0.0, 1.0]$ is a linear mixing coefficient rather than a fraction of response variance, $\bar{x}_{\text{delayed}}(t)$ is the continuum driver convolved with a top-hat transfer function of width 5 d centered at delay τ_{true}, and $\eta_{\text{indep}}(t)$ is an independent red-noise process. The simulations use the exact adopted observation dates and reported uncertainties. We evaluate $A \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and $\tau_{\text{true}} \in \{10, 20, 30\}$ d with 200 realizations per injected cell. Two independent sets of 200 null realizations calibrate and evaluate the threshold, giving 3400 central-grid realizations.

Under the pure null hypothesis, the empirical 95th percentile of the calibration maximum MI is 0.2009 bits. Fifteen of 200 independent evaluation realizations exceed it, giving FPR = 0.075 with a 95% Wilson interval of 0.046–0.120. The larger evaluation sample

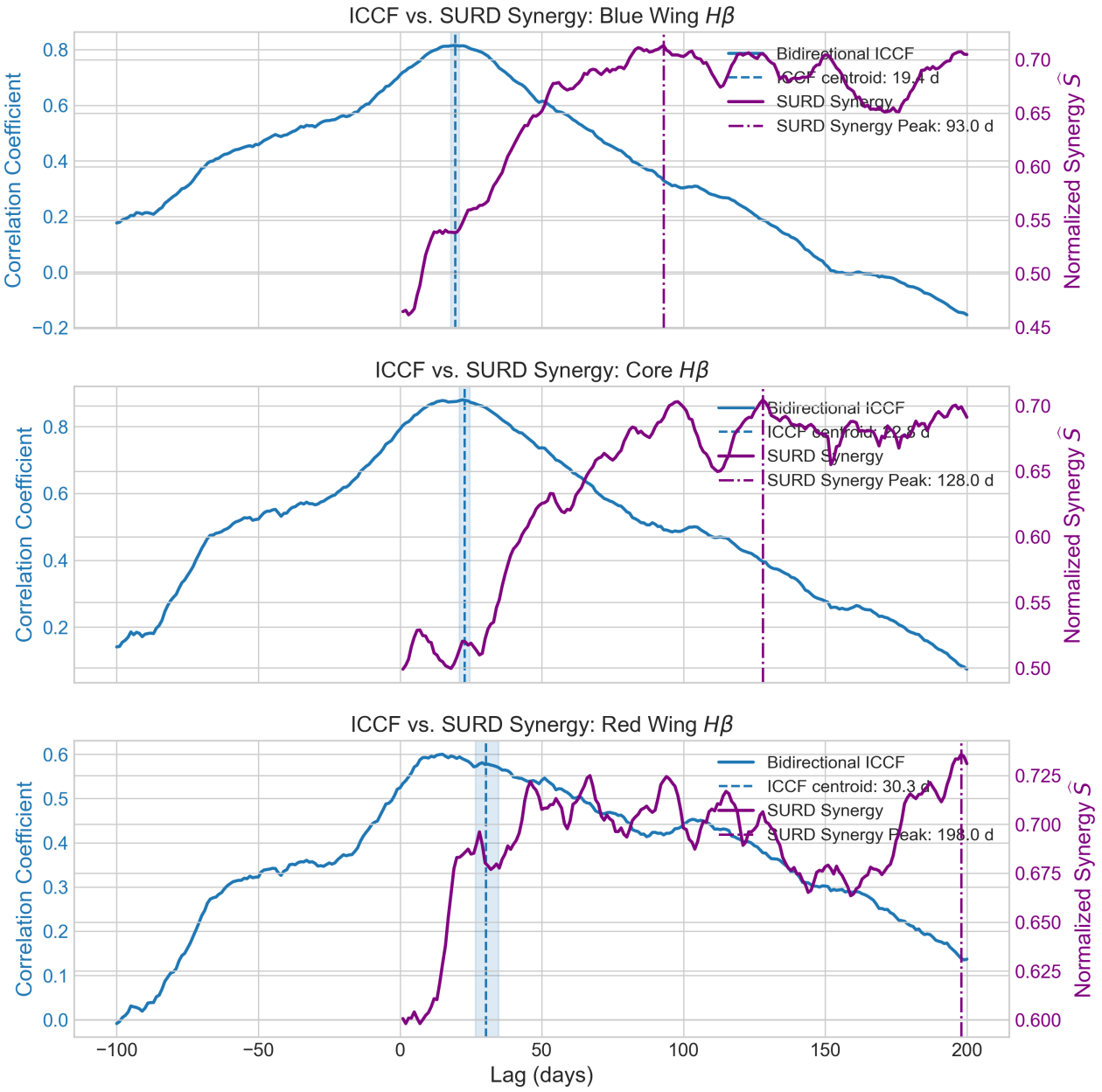


Figure 4. Native-cadence bidirectional ICCF (blue) and normalized three-predictor SURD total synergy (purple). Blue dashed lines and bands show the FR/RSS centroid median and 16th–84th percentiles; purple dash-dotted lines mark the descriptive SURD maxima. The quantities have different estimands and the SURD maxima are globally non-significant.

Target Component	Method	Peak Lag (d)	Peak MI (bits)	Null 95% (bits)	p_{global}	Leak Minimum (d)
Blue Wing Hβ	Gap-limited	16	0.473	0.425	0.009	16 (0.762)
Core Hβ	Gap-limited	15	0.719	0.494	0.001	15 (0.637)
Red Wing Hβ	Gap-limited	23	0.326	0.436	0.228	23 (0.834)
Blue Wing Hβ	Daily	20	0.446	–	–	20 (0.776)
Core Hβ	Daily	16	0.660	–	–	16 (0.660)
Red Wing Hβ	Daily	13	0.265	–	–	13 (0.867)

Table 5. One-predictor continuum baseline lag recovery and significance calibration across 201 lags (0–200 d). Statistical significance is evaluated against $N = 999$ circular-shift surrogates. Each global value controls the lag scan for one target. Benjamini–Hochberg adjustment across the three target-level tests gives $q = 0.0135$, 0.003, and 0.228 for Blue, Core, and Red.

Amplitude A	True Lag τ (d)	Detection Power	Recovery ≤ 3 d	Lag Bias (d)	Lag RMSE (d)	Mean Peak MI (bits)
0.0 (Null)	–	0.075	–	–	–	0.119
0.2	10	0.030	0.155	+18.98	27.98	0.138
0.2	20	0.050	0.185	+12.94	23.38	0.131
0.2	30	0.065	0.130	+4.84	19.66	0.139
0.4	10	0.445	0.640	+3.32	11.58	0.248
0.4	20	0.415	0.585	+2.08	10.77	0.249
0.4	30	0.415	0.575	+1.32	10.16	0.247
0.6	10	0.965	0.975	+0.13	1.54	0.528
0.6	20	0.945	0.950	−0.14	1.74	0.519
0.6	30	0.970	0.975	+0.02	1.64	0.541
0.8	10	1.00	1.00	−0.06	0.89	0.872
0.8	20	1.00	1.00	−0.08	0.89	0.897
0.8	30	1.00	1.00	−0.07	0.82	0.883
1.0	10	1.00	1.00	−0.09	0.69	1.016
1.0	20	1.00	1.00	−0.01	0.69	1.028
1.0	30	1.00	1.00	−0.02	0.78	1.035

Table 6. Detection power and lag recovery across the central grid. Independent null samples calibrate and evaluate the threshold. The false positive estimate is 0.075, with a 95% Wilson interval from 0.046 to 0.120.

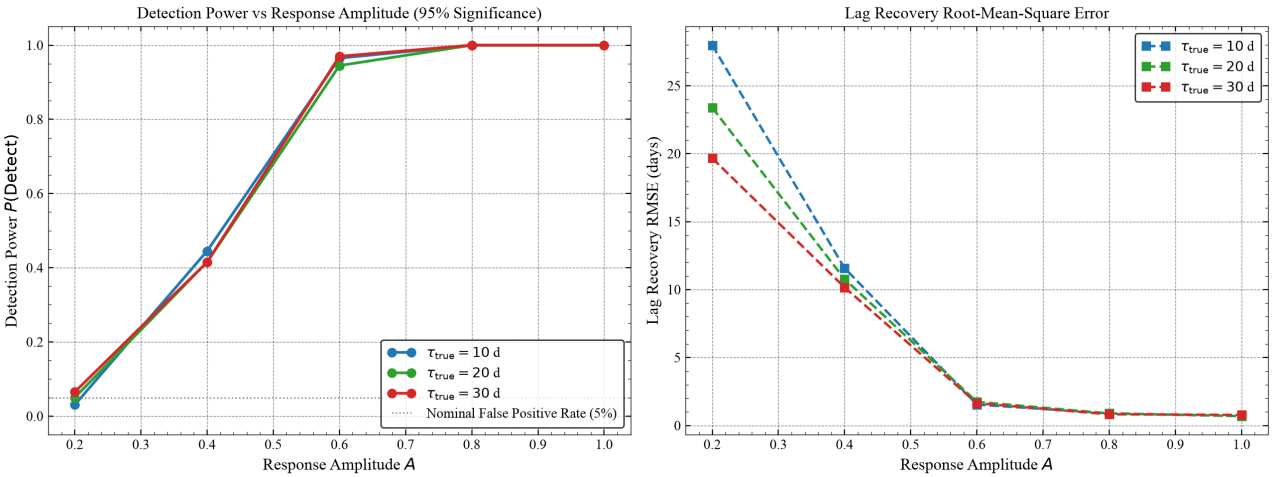


Figure 5. Detection power and lag RMSE as a function of the linear mixing coefficient A. At $A = 0.6$, power ranges from 0.945 to 0.970, while its 95% Wilson lower bounds range from 0.904 to 0.936.

narrows the interval, but the simulator remains a sensitivity model rather than a universal detection boundary.

Table 6 and Figure 5 reveal a sharp detection boundary:

(i) **Sub-threshold Regime ($A \leq 0.2$):** Detection power is negligible (2%–8%), and lag RMSE is large (19–31 d). The recovered peaks are scattered uniformly across the scan window by sample noise and interpolation trends.

(ii) **Intermediate Transition ($A = 0.4$):** Detection power rises to 36%–44%, with 54%–68% of realizations locating the lag within ±3 d. Lag RMSE drops to 10–16 d.

(iii) **High-response regime ($A \geq 0.6$):** At $A = 0.6$, power is 0.945–0.970 and lag RMSE is 1.54–1.74 d. At $A \geq 0.8$, every central-grid realization is detected, although finite-simulation intervals remain necessary.

(iv) **Asymptotic Precision ($A \geq 0.8$):** Detection power reaches 100%, with lag RMSE < 1.1 d and mean peak mutual information exceeding 0.87 bits.

This benchmark provides an exploratory sensitivity envelope for the stipulated simulator. One-factor runs over response widths of 2 and 10 d, process timescales of 30 and 100 d, and noise scales of 0.5 and 2 show material model dependence. The 100 d process gives only 0.84 power at $A = 0.6$ for all three delays. The central grid is now sufficiently sampled for this release, but the detection boundary remains model dependent.

4.7 Testing Incremental Velocity Information and Kinematic Coupling

In velocity-resolved reverberation mapping, an essential astrophysical question is whether different line components drive or predict one another—for example, whether the high-velocity wings transfer gas

into the low-velocity core, or whether an asymmetric outflow/inflow creates a directional lag between the blue and red wings (Xi et al. 2025). In an information-theoretic framework, this requires testing whether a candidate velocity component $X_{\rm wing}(t)$ provides *incremental predictive information* about another component $Y(t+\tau)$ after conditioning on both the driving continuum $F_{5100}(t)$ and the target's own red-noise history $Y(t)$:

$$H_0: \quad I(Y(t+\tau); X_{\rm wing}(t) \mid F_{5100}(t), Y(t)) = 0. \tag{20}$$

If H_0 holds, the observed correlations between velocity components are fully explained by their common response to the central continuum driver and their intrinsic temporal autocorrelation, leaving no evidence for directional kinematic transport.

We test four directional configurations:

(i) **Blue → Core:** Testing whether blue-wing gas predicts future core emission.
(ii) **Red → Core:** Testing whether red-wing gas predicts future core emission.
(iii) **Blue → Red (Outflow):** Testing whether the blueshifted wing leads and predicts the redshifted wing.
(iv) **Red → Blue (Inflow):** Testing whether the redshifted wing leads and predicts the blueshifted wing.

We use a discrete 3-bin quantile estimator and a KSG estimator with $k = 5$. Every KSG marginal is standardized before the scale-sensitive Chebyshev neighbour search. Significance uses $N = 100$ circular-shift surrogates in which the candidate component is rolled while the target, continuum, and target history remain fixed. Native gaps longer than 60 d define the primary support, and tuples cannot cross their boundary. As a sensitivity check, continuous neighbour counts were recomputed after excluding temporal neighbours within 1, 3, and 5 d. A separate four block contiguous date split was used for blocked prediction validation. Only two blocks had sufficient strict overlap at the 15 d evaluation lag, and both showed negative incremental R^2 values of −0.047 and −0.033.

Table 7 and Figure 6 show that no tested velocity wing has significant incremental information under the declared circular-shift null. This is evidence of nondetection, not proof that physical transport is absent:

- Blue → Core: discrete $p_{\rm global} = 0.911$, KNN $p_{\rm global} = 1.000$.
- Red → Core: discrete $p_{\rm global} = 1.000$, KNN $p_{\rm global} = 0.644$.
- Outflow (Blue → Red): discrete $p_{\rm global} = 0.832$, KNN $p_{\rm global} = 0.782$.
- Inflow (Red → Blue): discrete $p_{\rm global} = 0.941$, KNN $p_{\rm global} = 0.525$.

The blocked predictive diagnostic gives negative incremental R^2 in both usable held out blocks for the Core target and Blue candidate. Temporal neighbour exclusion changes the continuous estimator magnitude and therefore confirms estimator sensitivity, not a new detection. We conclude only that this aggregate campaign yields no significant directional CMI under the tested null. It neither detects nor refutes physical gas transport.

4.8 Unequal Predictor Lag Configurations

Equation 3.18 of the SURD formulation permits a distinct delay for every predictor. We evaluate common-lag, symmetric virial, blue-leading, and red-leading response-phase configurations for every target in two four-variable sets: continuum plus the three line components, and integrated Hβ plus the three line components. Predictor lags are defined relative to the response phase assigned to the selected target, so signed offsets are retained rather than silently clipping physically reversed orderings.

The corrected configurations are not equivalent to the common-lag control. Nevertheless, their synergy maxima remain broad and occur mainly near the upper portion of the 60 d base-lag scan, while leak minima are generally prompt. Without a pre-specified configuration family and matched null distribution, these differences cannot identify virial, inflow, or outflow geometry.

4.9 Synthetic Benchmark Demonstrations

To test the mathematical behavior of the histogram-based SURD pipeline under realistic observational conditions, we construct four synthetic benchmark models. The continuous driver signals are generated as damped random walk (DRW) processes on a 1-day grid, with pre-history included so that applying a delay does not wrap the end of a realization onto its beginning. The target signals are then constructed with known delays and couplings:

(i) **Case 1: Single Driver.** The target responds to only one driver with a delay of 15.0 days.
(ii) **Case 2: Redundant Proxies.** The second driver is a noisy copy of the first. The target responds to the first driver with a 15.0-day delay.
(iii) **Case 3: Joint Nonlinear Drivers.** The target depends on the product of two independent drivers delayed by 15.0 days.
(iv) **Case 4: Two Independent Delayed Drivers.** The target is the additive response $0.5S_1(t-10)+0.5S_2(t-20)$, where S_1 and S_2 are independent DRW processes.

To match the observational footprint of the real campaign, the first predictor is sampled at the 557 continuum measurements and the second predictor and target are sampled at the 242 spectroscopic measurements (224 distinct dates in the adopted sample). Measurements sharing a date are averaged before interpolation. Gaussian noise is scaled by each channel's measured uncertainty relative to its observed flux standard deviation, and the sampled curves are interpolated back onto the 1-day grid. We run 100 independent Monte Carlo realizations, scan lags from 1 to 40 d, and record both lag recovery and whether the intended information atom is the largest atom at the true lag.

Figure 8 and Table 8 separate simple positive controls from a harder two-delay stress test. Cases 1–3 recover the 15-day lag within ±5 d in every realization, with mean biases smaller than 0.1 d. Their intended atoms are largest at the true lag in 100%, 100%, and 99% of realizations, respectively. The additive two-delay case is substantially less reliable: the U_1 and U_2 diagnostics recover their associated lags within ±5 d in only 67% and 71% of realizations, and are almost never the largest atom at the true lag. Thus, the estimator passes isolated-mechanism controls but does not reliably separate overlapping, autocorrelated responses.

The failure in Case 4 is visible in the median decomposition: broad synergy and redundancy exceed the individual unique terms even though the generating equation is additive. This is an important boundary on interpretation. A peak in a selected atom may locate a delay under a simple controlled mechanism, but the identity and amplitude of that atom cannot be mapped directly onto a physical coupling when multiple autocorrelated responses overlap. None of these prompt-lag tests supports interpreting the real 93–198 d maxima as displaced versions of 10–20 d reverberation delays.

The positive controls establish that the implementation can recover

Target Y	Candidate Wing $X_{\rm wing}$	Peak Lag (d)	Peak CMI (bits)	Null 95% (bits)	$p_{\rm global}$	$\Delta R^2_{\rm held\text{-}out}$
Core Hβ	Blue Wing (discrete)	96	0.218	0.438	0.911	−0.031
Core Hβ	Blue Wing (KNN $k = 5$)	43	0.156	0.322	1.000	–
Core Hβ	Red Wing (discrete)	100	0.192	0.378	1.000	−0.058
Core Hβ	Red Wing (KNN $k = 5$)	52	0.233	0.306	0.644	–
Red Wing Hβ	Blue Wing (Outflow, discrete)	39	0.211	0.334	0.832	−0.187
Red Wing Hβ	Blue Wing (Outflow, KNN)	30	0.128	0.214	0.782	–
Blue Wing Hβ	Red Wing (Inflow, discrete)	34	0.253	0.419	0.941	−0.102
Blue Wing Hβ	Red Wing (Inflow, KNN)	97	0.281	0.338	0.525	–

Table 7. Tests of incremental predictive information conditioned on the continuum driver and target history. Global values use 100 circular-shift surrogates. Continuous variables are standardized before KSG neighbour searches. The final column is the sample-count-weighted change in predictive R^2 at the pre-specified 15 d lag across the two native-gap-defined segments used by the primary diagnostic. A separate four block contiguous date sensitivity check is reported in the text.

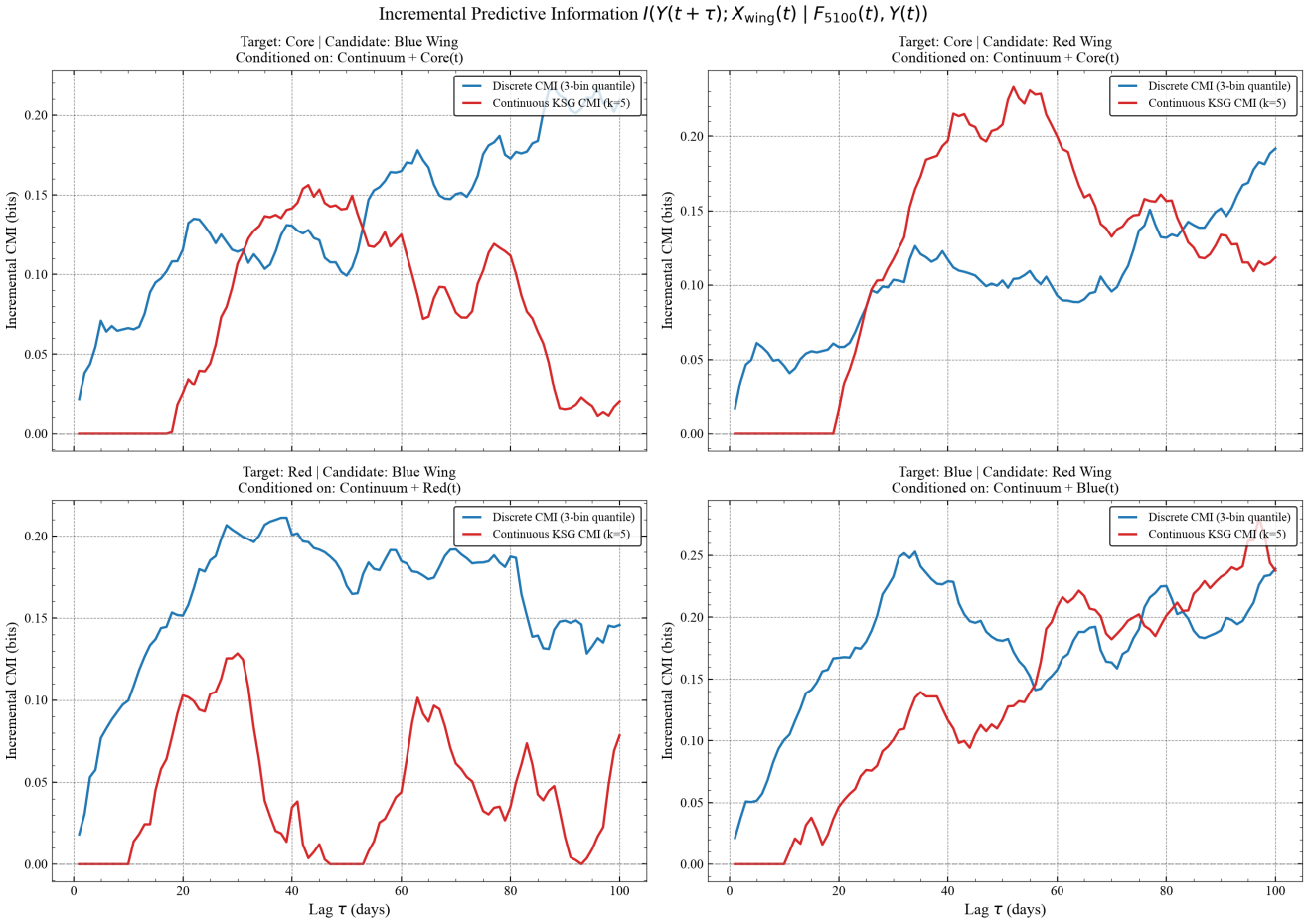


Figure 6. Observed conditional mutual information scans for discrete quantile histograms (blue) and standardized continuous KSG estimation (red). The plotted curves are descriptive; scan-wide significance is obtained from surrogate maxima and reported in Table 7. Every tested directional configuration is non-significant under both estimators.

simple isolated mechanisms under the observing window. The two-delay stress test shows that this success does not generalize to a more ambiguous multicomponent response, for which both lag recovery and atom identity degrade.

4.10 Matched Negative Control and Ablation

We next isolate an estimator-level null with no cross-channel coupling. For each of 200 realizations, we generate four independent Ornstein–Uhlenbeck red-noise series on the same 1744-d window and analyse the core analogue from the other three channels using the primary eight-bin, three-predictor, normalized 1–200 d scan. We compare (i) complete daily series, (ii) values retained only at exact observed continuum and spectroscopic epochs, and (iii) those observed values linearly interpolated back to the daily grid. At each stage we record normalized total synergy, absolute total synergy, joint mutual information, and the scan maximum.

Figure 9 shows that large normalized values and scan-selected

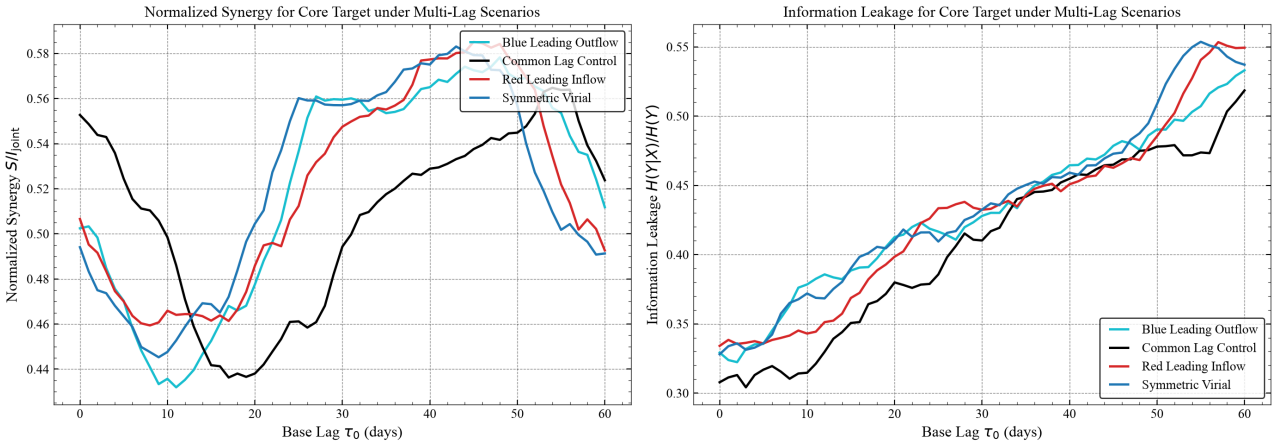


Figure 7. Core-target normalized synergy and information leakage for four unequal-lag configurations. The complete saved table includes every target in both four-variable sets. These curves are descriptive because an unequal-lag maximum-statistic null has not yet been calibrated.

Case Model	True Lag	Diagnostic	Recovered Lag	Bias	Within ±5 d	Atom Largest
Case 1: Single Driver	15	U_1	15.0 ± 0.8	0.0	1.00	1.00
Case 2: Redundant Proxies	15	R_{12}	15.0 ± 0.8	+0.0	1.00	1.00
Case 3: Joint Nonlinear	15	S_{12}	15.0 ± 1.2	−0.0	1.00	0.99
Case 4: Driver 1	10	U_1	8.4 ± 8.4	−1.6	0.67	0.01
Case 4: Driver 2	20	U_2	24.9 ± 5.8	+4.9	0.71	0.00

Table 8. Lag and information-atom recovery across 100 Monte Carlo realizations under the NGC 5548 sampling footprint. Recovered lags are means and sample standard deviations in days. "Atom Largest" is the fraction of realizations in which the designated atom exceeds the other three two-predictor atoms at the true lag.

maxima occur despite complete independence. Their amplitude and location change under masking and interpolation, while absolute synergy and joint mutual information behave differently from their ratio. The exact-epoch calculation has only 11–40 usable matched tuples, depending on lag, and is therefore itself too sparse for a quantitative four-dimensional histogram estimate. This ablation does not identify one unique cause. It demonstrates the narrower claim required here: four-dimensional histogram sparsity, red-noise structure, the observing window, interpolation, and normalization are jointly sufficient to reproduce unstable false maxima. Interpolation alone is not assigned a causal effect.

4.11 Target-History Conditioning and Autocorrelation

A key challenge in information-theoretic analyses of astronomical time series is distinguishing true cross-component information flow from the target signal's own autocorrelation (red-noise memory). If a target component (such as Core Hβ) is strongly autocorrelated on short timescales, the apparent synergy between predictors could simply reflect the target's memory rather than direct physical coupling from the driving continuum.

To address this, we perform a target-history conditioned SURD scan. We decompose the conditional mutual information:

$$I(Y(t+\tau); X_1(t), X_2(t) \mid X_3(t)) \tag{21}$$

where $Y = \text{Core}(t+\tau)$ is the future state of the target Core, $X_1 = \text{Continuum}(t)$ and $X_2 = \text{Blue Wing}(t)$ are the predictors, and $X_3 = \text{Core}(t)$ is the target's own past state. This conditional Partial Information Decomposition (conditional PID) is calculated by binning the joint 4D space, extracting the 3D slices for each bin of X_3, running the standard SURD decomposition on each slice, and taking the probability-weighted sum of the results.

The results of this comparison are shown in Figure 10. Because the conditional surrogate suite was generated only over 1–120 d, we restrict the comparison to that same window. Within this tested range, the unconditioned synergy curve peaks at **9 days** (0.316 bits), reflecting strong short-term autocorrelation of the light curve, while the history-conditioned curve reaches its local maximum at **73 days** (0.309 bits).

However, surrogate significance tests and bin occupancy analysis show that this conditional peak is not statistically significant:

(i) **Conditional Surrogate Significance:** We run 100 circular-shift surrogates for the conditional scan to estimate the global null distribution of the max-statistic. The real peak conditional synergy (0.3092 bits) is well below the median of the global null max-statistic (0.4043 bits), yielding a global p-value of **0.9802**. This indicates that the conditioned peak is statistically indistinguishable from random noise.

(ii) **High-Dimensional Sample Sparsity:** The conditional decomposition requires binning a 4D joint space. The daily strict-overlap grid has $N = 1744$, but a lag τ uses $N_{\text{eff}}(\tau) = 1744 - \tau$ tuples. Thus the 73-d conditional maximum has $N_{\text{eff}} = 1671$. For $n_{\text{bins}} = 6$, the $6^4 = 1296$ cells are 89.20% empty (1156 cells), with only 11.9 samples per occupied cell on average. This severe sparsity biases the histogram information estimates.

We then repeat the complete 1–120 d conditional scan with 3–8 bins using both equal-width and quantile discretization. For every setting, we generate 49 additional circular-shift realizations and compare the observed maximum with the distribution of surrogate maxima. Ta-

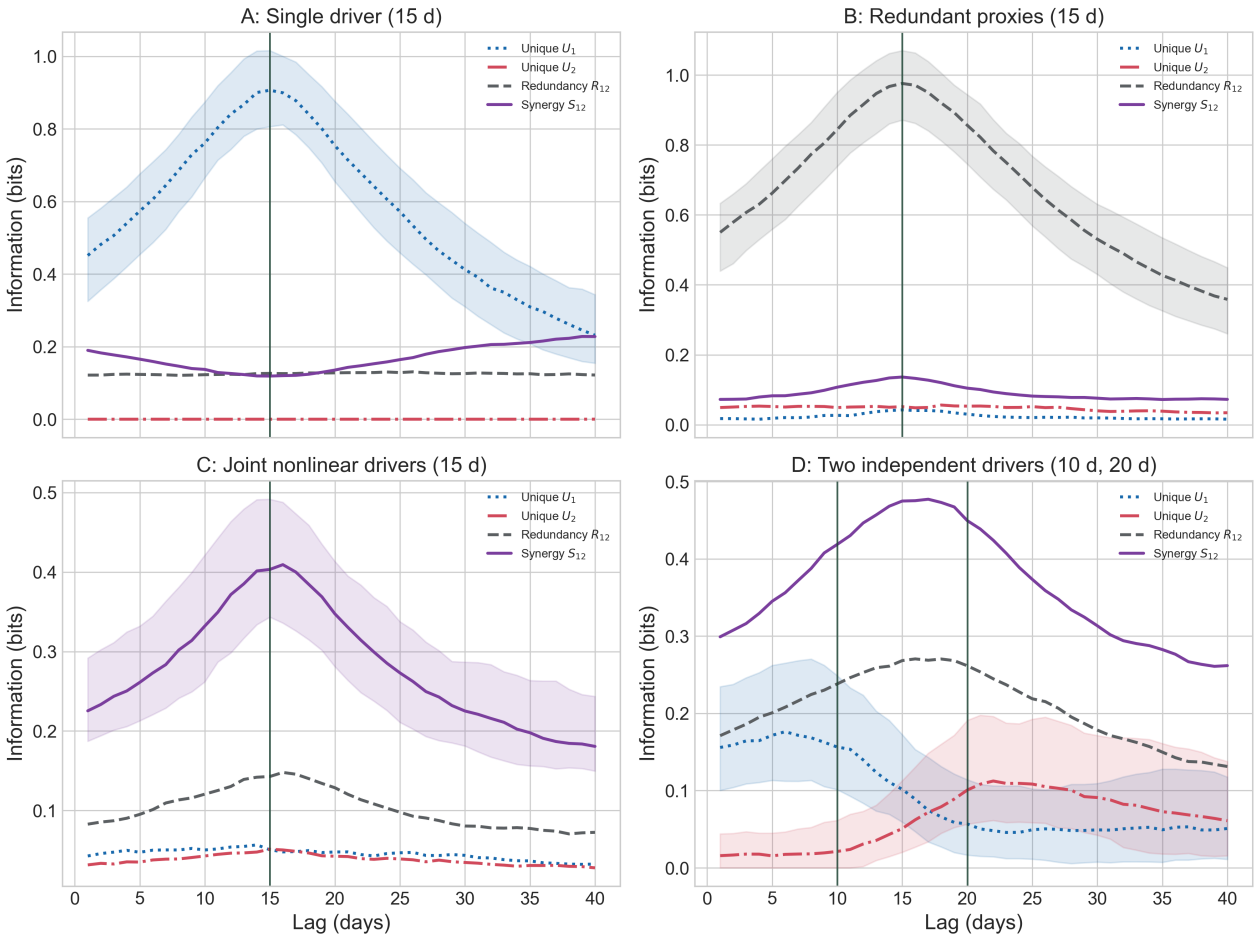


Figure 8. Median SURD decompositions over 100 synthetic realizations sampled under the NGC 5548 observing footprint. Shading spans the 16th–84th percentiles for the designated diagnostic atom (two atoms in Case 4); vertical lines mark the true delays. Cases 1–3 recover their isolated generating mechanisms, whereas the additive two-delay stress test is dominated by broad synergy and redundancy.

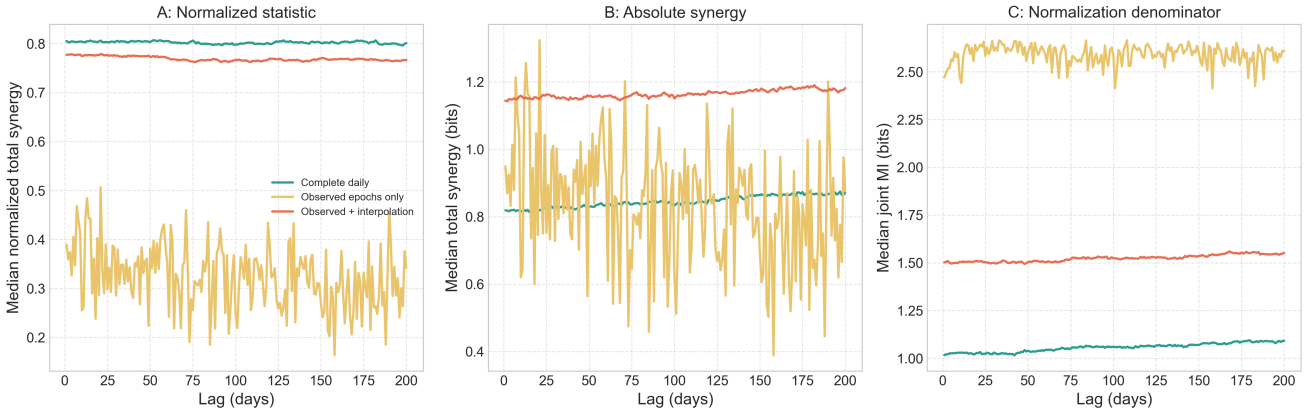


Figure 9. Matched three-predictor negative-control ablation over 200 independent red-noise realizations. Curves show medians for complete daily data, exact observed epochs, and observed epochs followed by daily interpolation. The panels separate normalized total synergy, absolute total synergy, and its joint-MI normalization denominator. The observed-only estimate is shown diagnostically but is severely undersampled.

ble 9 and Figure 11 show that the inferred peak is not stable: equal-width peaks span 37–97 d and quantile-bin peaks span 71–102 d. The estimated peak amplitude also grows strongly with bin count as the joint histogram becomes sparser. Despite these changes, every auxiliary global test is non-significant, with empirical $p_{\rm global}$ values from 0.64 to 1.00. The 49-surrogate auxiliary tests have a p-value resolution of 0.02 and are intended as sensitivity checks; the primary six-bin significance result remains the 100-surrogate value $p_{\rm global} = 0.9802$ quoted above.

Consequently, within the tested 1–120 d window, the conditional peak is better interpreted as a discretization- and finite-sample-dependent feature than as a resolved physical coupling. The robust result is the absence of global significance, not the location of any individual maximum.

4.12 Dimensionality Limits and Estimator Scaling

The severe sparsity encountered in the target-history conditioned scan (Section 4.11) highlights a fundamental limitation of histogram-based information-theoretic estimators: the exponential scaling of the joint probability cell count with the number of variables (the "curse of dimensionality").

To quantify this effect, we record occupancy at each primary maximum. The $8^4 = 4096$-cell histograms use $N_{\rm eff} = 1629$, 1546, 1616, and 1651 tuples at the continuum (115 d), red (198 d), core (128 d), and blue (93 d) maxima. Their empty-cell fractions are 93.82%, 93.65%, 93.19%, and 93.04%, respectively. Thus the effective sample count and occupancy are explicitly lag-dependent, and the four-dimensional histogram estimates remain exploratory.

Adding target history as a fourth predictor with eight bins would require $8^5 = 32,768$ cells for at most 1743 lagged tuples, or fewer than 0.054 samples per cell even before accounting for nonuniform occupancy. We therefore do not attempt that histogram decomposition.

Consequently, the 3-predictor round-robin scans can at most identify exploratory information profiles, while full 4-predictor joint reconstructions are out of reach using standard histograms. To test whether the principal lag and significance conclusions depend on histogram sparsity, we perform the continuous-estimator cross-check below.

4.13 Continuous-Estimator Cross-Check

We estimate total mutual information between each future target and the other three present channels using the Kraskov–Stögbauer–Grassberger (KSG) k-nearest-neighbour estimator (Kraskov et al. 2004). For samples $i = 1, \ldots, N$, the estimator uses the distance to each sample's kth neighbour in the joint space and counts neighbours $n_x(i)$ and $n_y(i)$ in the corresponding marginal spaces,

$$\widehat{I}_{\rm KSG} = \psi(k) + \psi(N) - \left\langle \psi[n_x(i)+1] + \psi[n_y(i)+1] \right\rangle_i , \qquad (22)$$

where ψ is the digamma function. We use the Chebyshev metric, add deterministic jitter of 10^{-10} times the marginal scale to break exact interpolation ties, and repeat the calculation for $k = 3$, 5, and 10. We also estimate $I(Y; X_1, X_2 \mid Y_{\rm past})$ with the nearest-neighbour conditional mutual-information estimator of Frenzel & Pompe (2007). Significance is evaluated with 999 independent circular-shift realizations using the same global maximum-statistic procedure as the histogram analysis.

Table 10 reports the central $k = 5$ results; $k = 3$ and 10 give the same qualitative conclusion. Total KNN mutual information identifies globally significant prompt maxima at 1 d for the three line targets ($p_{\rm global} = 0.001$ for $k = 5$). In contrast, none of the maxima restricted to 61–200 d is significant: across all three k choices, $p_{\rm global} = 0.167$–0.271 for the Blue Wing, 0.418–0.466 for the Core, 0.592–0.888 for the Red Wing, and 0.861–0.911 for the Continuum. Values evaluated directly at the histogram-SURD maxima are also non-significant ($p = 0.194$–0.907). Finally, the history-conditioned scan is globally non-significant for every neighbourhood size ($p_{\rm global} = 0.977$–0.985), and its value at 73 d is pointwise non-significant ($p = 0.915$ for $k = 5$).

This cross-check does not reproduce the SURD atom decomposition and is not presented as an alternative unique–redundant–synergistic partition. It establishes a narrower but important result: a continuous estimator independently recovers strong prompt multivariate dependence while providing no evidence that the reported long-lag SURD maxima represent significant predictive dependence.

4.14 Independent-Object Check with Mrk 817

We next test whether the estimator-sensitivity seen in NGC 5548 is specific to that object. We use the public AGN STORM 2 HST/COS light curves of Mrk 817: 165 synchronized visits spanning 456.6 d, with a median cadence of 2.0 d, for the 1180-Å continuum and Lyα, N V, Si IV, C IV, and He II emission lines (Homayouni et al. 2023; Plesha et al. 2023). Published rest-frame continuum–line delays range from 8.2 to 15.5 d (Homayouni et al. 2023).

Because this campaign contains gaps of 36 and 51 d, we do not interpolate it onto a daily grid. At each integer observed-frame lag from 0 to 40 d, continuum and line measurements are instead paired one-to-one when their actual separation is within 1.1 d of the requested lag. We estimate bivariate KSG mutual information for $k = 3$, 5, and 10, and compare each curve with 999 circular shifts of the corresponding line fluxes. We report a line-wise maximum-statistic p-value, a stricter family-wise value based on the largest statistic across all five lines and lags, and a separate five-line family-wise test at the externally specified published delays. Pairing tolerances of 0.75 and 1.5 d provide a sensitivity check.

Table 11 gives the central $k = 5$ results. At the published delays, Lyα and C IV remain significant after the five-line correction ($p = 0.027$ and 0.001), whereas N V, Si IV, and He II do not ($p = 0.053$, 0.065, and 0.993). The unrestricted scan is less stable: only N V narrowly survives the five-line, 0–40 d correction for $k = 5$ ($p = 0.048$), but its maximizing lag is 5 d rather than the published 15.5 d and moves between 5 and 15 d as the pairing tolerance and k change. The other four $k = 5$ maxima have family-wise $p = 0.171$–1.000. Thus the independent object contains real information near some externally measured delays, but maximizing an unrestricted information curve does not yield a generally stable lag estimator. This supports the methodological conclusion from NGC 5548 without extending the SURD atom decomposition to a sample too small for a reliable multidimensional histogram.

5 INTERPRETATION AND METHODOLOGICAL IMPLICATIONS

The results in Section 4 reveal a discrepancy between native-cadence ICCF centroid delays (approximately 19–30 d) and normalized three-predictor SURD maxima (93–198 d). We interpret that discrepancy in light of the global nulls, synthetic controls, estimator sensitivity, and known BLR context.

Binning	Bins	Peak Lag (d)	Peak S_{12} (bits)	Empty Cells	$p_{\rm global}$
Equal width	3	62	0.080	58.0%	0.96
Equal width	4	37	0.146	77.7%	0.98
Equal width	5	74	0.223	83.8%	0.94
Equal width	6	73	0.309	89.2%	0.96
Equal width	7	60	0.408	92.3%	0.92
Equal width	8	97	0.467	94.5%	0.94
Quantile	3	71	0.146	29.6%	0.96
Quantile	4	85	0.320	55.1%	1.00
Quantile	5	92	0.428	70.4%	0.98
Quantile	6	92	0.572	80.9%	0.80
Quantile	7	97	0.679	87.1%	0.72
Quantile	8	102	0.725	91.0%	0.64

Table 9. Sensitivity of the core-target history-conditioned scan to histogram construction. Empty-cell fractions are evaluated at each setting's observed maximum. Empirical global p-values use 49 auxiliary circular-shift surrogate maxima per setting. None is significant at $\alpha = 0.05$.

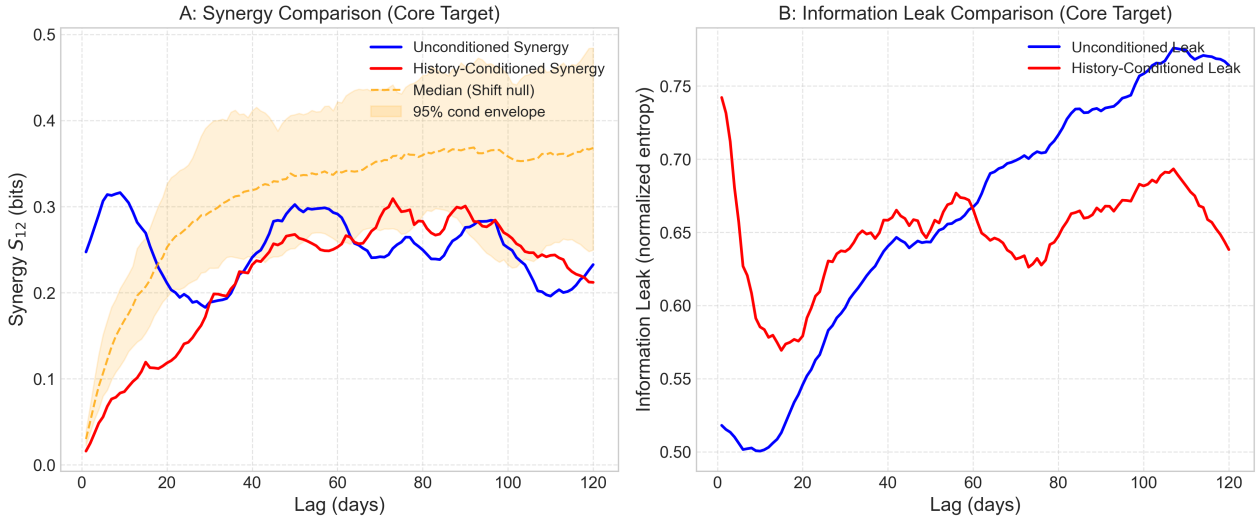


Figure 10. Comparison of the unconditioned (blue) and target-history conditioned (red) SURD synergy (left) and information leak (right) curves for the Core Hβ target over the 1–120 d window for which conditional surrogates were computed. The orange shaded band and dashed line represent the pointwise 95% circular-shift surrogate envelope and median at each lag for the conditioned scan. The global max-statistic test compares the observed peak to the distribution of per-realization surrogate maxima (whose median is 0.4043 bits, with the 95th percentile higher) and is not represented by a single pointwise threshold curve. Within this tested window, conditioning on the target's own past shifts the local synergy peak to 73 days, but the peak is statistically insignificant ($p_{\rm global} = 0.9802$) under this null and does not establish physical coupling.

Target	Peak Lag	p_{1--200}	Long-Lag Peak	$p_{61--200}$	SURD Peak	$p_{\rm at\ SURD}$
Continuum	94	0.942	94	0.861	115	0.862
Blue Wing	1	0.001	168	0.167	93	0.216
Core	1	0.001	61	0.418	128	0.493
Red Wing	1	0.001	164	0.787	198	0.681
Core conditional	77	0.977	–	–	73	0.915

Table 10. Continuous KNN estimator cross-check for $k = 5$ using 999 circular-shift surrogates. Peak lags are in days. The first global p-value covers the full scan (1–200 d for total MI and 1–120 d for conditional MI); the long-lag test is restricted to 61–200 d. The final column is pointwise at the corresponding histogram-SURD maximum and is not a global test.

5.1 The Multi-Component Broad-Line Region

The Hβ reverberation lag of NGC 5548 is historically reported near tens of days (e.g., 15.6 ± 0.5 days during the high-cadence AGN STORM campaign (Pei et al. 2017)). Our blue and core ICCF centroid distributions are prompt, while the red-wing function has a 15-d peak but a broad shoulder that moves its centroid to approximately 30 d. These summaries support a prompt reverberating response but do not by themselves determine BLR geometry or a velocity ordering. More detailed line-shape and delay-profile modelling motivates multicomponent descriptions (Long & Dexter 2025).

Our one-predictor continuum baseline identifies prompt peaks at

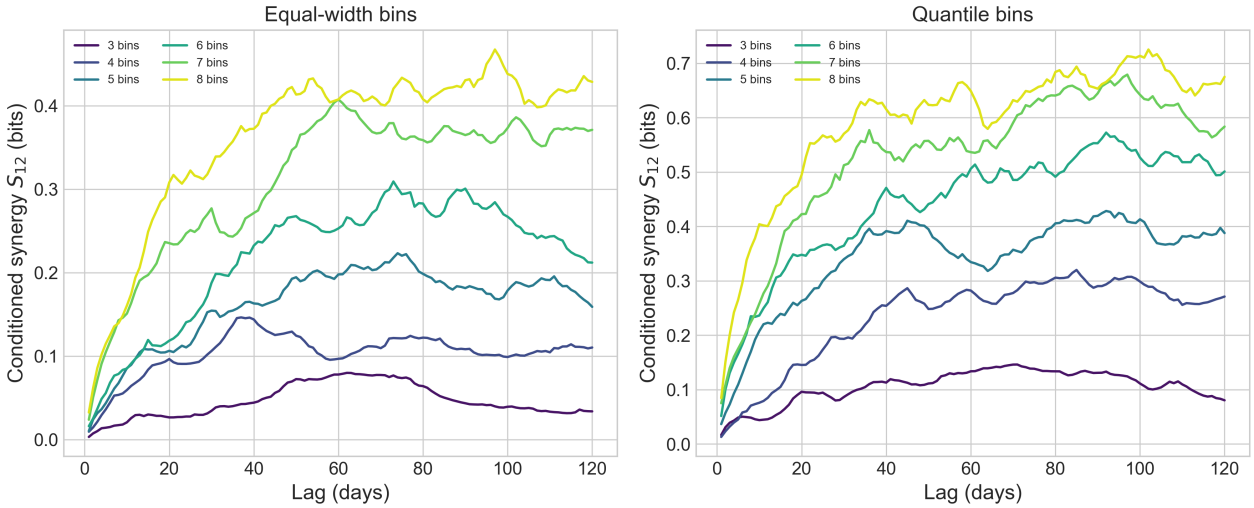


Figure 11. History-conditioned core-target synergy over 1–120 d for 3–8 equal-width bins (left) and quantile bins (right). Both peak amplitude and peak location vary with discretization; increasing the number of bins systematically inflates the estimated synergy as occupancy declines.

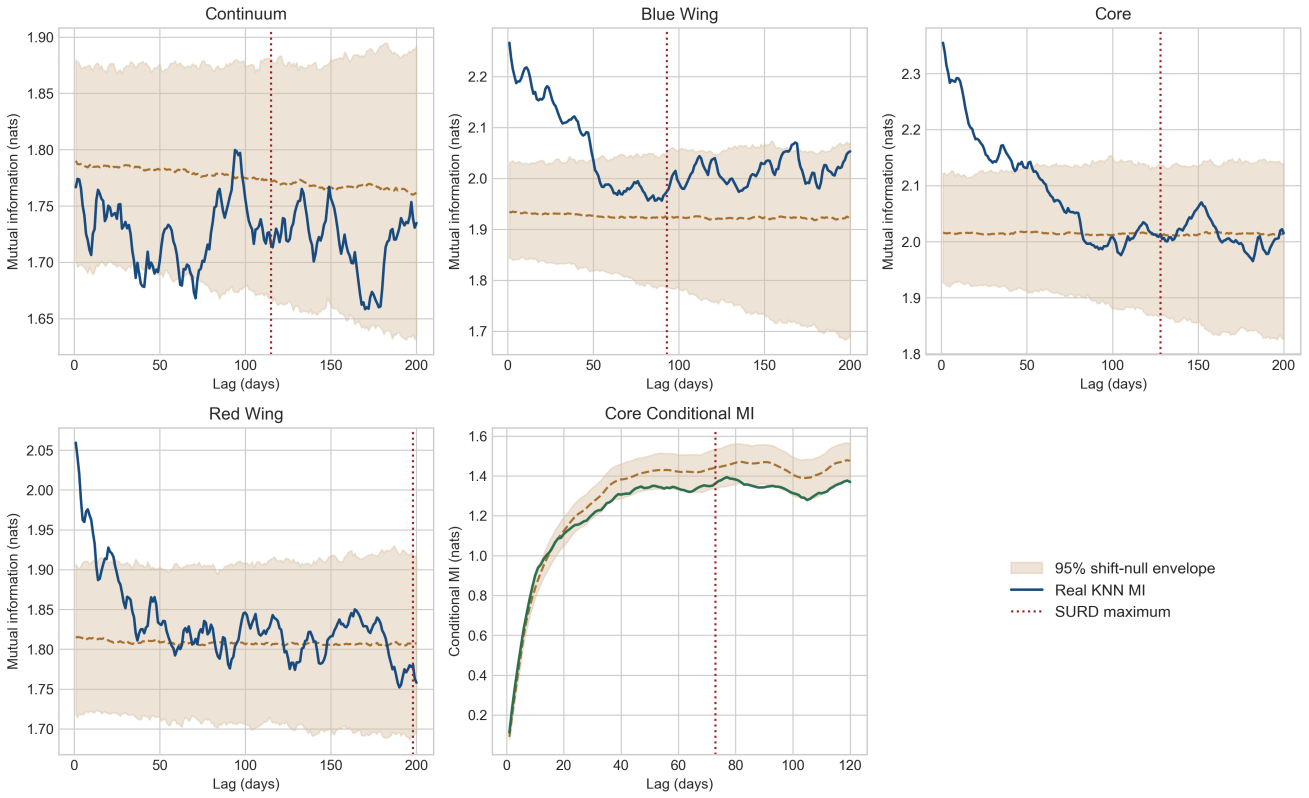


Figure 12. KNN mutual-information cross-check for $k = 5$. Blue or green curves show the real total or conditional mutual information, tan bands show pointwise 95% circular-shift envelopes, and dotted red lines mark the corresponding histogram-SURD maxima. Prompt dependence is visible for the three line targets, whereas the long-lag SURD maxima and the conditioned 73-day maximum lie within their null distributions.

16 d for the Blue Wing and 15 d for the Line Core. With 999 surrogates, their lag-scan global values are 0.009 and 0.001, and their three-target adjusted values are 0.0135 and 0.003. The long three-predictor maxima require interpretation against estimator and sampling effects rather than as physical delays:

(i) **Extended Geometry:** Extended geometry or non-linear photoionization reprocessing could in principle contribute to long-timescale dependence. However, the matched independent-red-noise control reproduces comparable maxima under the same dimensionality and estimator. The observed maxima therefore do not require a physical long-lag component.

(ii) **Non-linear Reprocessing:** Non-linear reprocessing cannot be

Line	MI peak	p_{line}	$p_{5\,\text{lines}}$	Published lag	$p_{\text{published},5\,\text{lines}}$
Lyα	14	0.137	0.424	10.4	0.027
N V	5	0.001	0.048	15.5	0.053
Si IV	9	0.080	0.876	8.2	0.065
C IV	19	0.085	0.171	11.8	0.001
He II	10	0.126	1.000	9.0	0.993

Table 11. Independent Mrk 817 KNN check for $k = 5$ using 999 circular-shift surrogates. MI-peak lags are observed-frame days; published lags are rest-frame days from Homayouni et al. (2023). The two maximum-statistic columns correct over 0–40 d for one line and all five lines, respectively. The final column corrects across the five externally specified literature lags.



Figure 13. Native-cadence KNN mutual-information scans for Mrk 817. Coloured curves show $k = 3$, 5, and 10; grey dashed curves are the pointwise 95% circular-shift envelopes for $k = 5$; dotted red lines mark the observed-frame equivalents of the published delays. Pointwise excursions are not equivalent to the scan-wide tests in Table 11.

inferred from peak amplitude alone because the matched fully independent controls have median normalized total synergy near 0.8, matching or exceeding the real-data amplitudes.

(iii) **Estimator and Sampling Effects:** Four-dimensional histogram sparsity, red-noise structure, the observing window, interpolation, and normalization are jointly sufficient to reproduce unstable maxima. The migration of the three-predictor maxima to long lags is an artifact of cross-line predictor collinearity and seasonal red-noise aliasing.

(iv) **Observational Sensitivity Boundaries:** In the central simulator, $A = 0.6$ gives power from 0.945 to 0.970. One-factor sensitivity runs show weaker performance for a 100 d process timescale, so A is not a universal threshold and is not a variance fraction.

5.2 Comparison with 2023 Reverberation Mapping and Kinematic Evolution

A key context for our results is the recent spectroscopic reverberation campaign of NGC 5548 presented by Xi et al. (2025) and the five seasons that precede it (Lu et al. 2022). The pooled window used here spans the five historical campaigns, whose individual Hβ lags in the compilation of Lu et al. (2022) range from 11.0 to 19.7 days. Xi et al. (2025) measure an integrated Hβ rest-frame delay of $15.6^{+2.6}_{-2.9}$ days. This is consistent with the prompt peak structure in our ICCFs and continuum baseline mutual information, although our velocity-resolved centroids are not measurements of the same integrated quantity.

However, Lu et al. (2022) and Xi et al. (2025) show that the velocity-resolved Hβ lag profile of NGC 5548 is highly nonstationary and changes substantially from season to season:

- **Virialized/Keplerian (e.g., 2015):** A symmetric, M-shaped lag

profile where the high-velocity wings respond faster than the low-velocity core ($|v| \uparrow \implies \tau \downarrow$), consistent with virial motion.

- **Inflow-like (e.g., 2019, 2021):** A red-leading-blue asymmetry where the redshifted wing responds earlier ($\tau_{\rm red} < \tau_{\rm blue}$), indicating radial inflow.
- **Outflow-like (e.g., 2020, 2023):** A blue-leading-red asymmetry ($\tau_{\rm blue} < \tau_{\rm red}$), indicating radial outflow or wind-driven gas.

Our incremental tests ask whether these kinematic patterns appear as persistent directional information over the historical aggregate campaign. No tested direction is significant under either estimator, with global values from 0.525 to 1.000. The native gap rule yields only two predictive validation segments, so their negative weighted changes in R^2 are diagnostics rather than decisive confirmation. These results do not test the later Xi et al. seasons and do not refute transient or season-specific transport.

5.3 Scope of the Information-Decomposition Framework

While ICCF and transfer-function fitting are designed to estimate characteristic response times, SURD asks how predictive information is distributed across a specified multivariate predictor set. Its unique, redundant, synergistic, and leak terms are descriptive information quantities; under the present sampling and estimator they must not be read as direct causal or geometric components.

The terms have the following operational interpretations:

- **Unique Information:** Information associated with one predictor that is not available from the other predictors included in that scan. Its value depends on the chosen predictor set and does not by itself establish a direct physical pathway.
- **Redundancy:** Predictive information available from more than one included predictor. A common driver is one possible source, but correlated sampling, autocorrelation, and interpolation can also contribute.
- **Synergy:** Predictive information available only from a joint predictor state under the decomposition. The synthetic stress test and real-data null tests show that a large estimated synergy is not sufficient evidence for multivariate physical coupling.
- **Predictive Leak ($\mathcal{L}$):** The conditional-entropy fraction left unexplained by the observed predictors. It combines genuinely missing influence with noise, stochasticity, discretization bias, and model or stationarity failure, so it cannot uniquely identify a hidden driver.

This framework supplements the question "what is the delay?" with "how is estimated predictive information distributed across the monitored components, and what fraction remains unexplained?"

For the present NGC 5548 observations, however, that richer description does not yield a demonstrated astrophysical advantage over established lag analyses. No unique, redundant, or synergistic feature is both globally significant and robust to the tested estimator and analysis choices. The prompt relationships are recovered more directly by ICCF and simpler information measures. SURD therefore contributes here as a diagnostic and validation framework, rather than as a source of additional physical knowledge about the BLR.

5.4 Interpretation of the Information Leak

The normalized information leak, defined as the ratio of conditional entropy to total target entropy:

$$\mathcal{L} = \frac{H(Y(t+\tau) \mid \mathbf{X}(t))}{H(Y(t+\tau))}, \tag{23}$$

measures the proportion of the target's future uncertainty that remains unexplained by the observed predictors. It is an information-theoretic entropy fraction, not a linear explained-variance statistic (like R^2), and it naturally incorporates finite-sample bias, discretization effects, observational noise, missing predictors, and stochasticity.

For the three line targets in the 3-predictor round-robin scan, the minimum leak occurs at 1 day (Red Wing), 9 days (Core), and 9 days (Blue Wing), with residual entropy fractions of approximately 0.42, 0.25, and 0.30, respectively. These minima are much shorter than the long-lag synergy maxima and are therefore consistent with a prompt predictive horizon. However, we do not treat the exact leak minimum as a standalone reverberation lag measurement: the red-wing minimum at 1 day, together with the continuum-target minimum at 164 days, shows that the leak surface is sensitive to predictor choice, normalization, and estimator bias. In contrast to the unconditioned 2-predictor scan, when we introduce the target's own history as a predictor (Section 4.11), the target's strong self-correlation can dominate the short-lag behavior unless it is explicitly conditioned out.

As the lag increases beyond these timescales, the information leak rises, representing a loss of predictive power as we move away from the predictive horizon. The remaining unexplained entropy (the leak floor) is likely caused by:

- **Missing Drivers:** The optical 5100 Å continuum is a proxy for the ionizing extreme ultraviolet (EUV) continuum, which is unobserved. The physical decoupling during events like the "reverberation holiday" introduces unexplained variance.
- **Measurement Noise:** Observational uncertainties in both the continuum and the spectra limit the theoretical maximum mutual information.
- **Discretization and Estimation Bias:** The use of histogram-based probability density estimators with finite binning introduces systematic sample-size bias that prevents the leak from reaching zero.

5.5 Limitations of the Current Study

While this work demonstrates the application of the SURD framework to velocity-resolved reverberation mapping, several key physical and methodological limitations must be kept in mind:

- **Primary Single-Object Focus:** The SURD decomposition is restricted to NGC 5548. The Mrk 817 calculation is an external KNN dependence check, not an independent replication of the atom partition, so population-level generalization remains premature.
- **Lack of Directly Observed Ionizing Continuum:** The optical 5100 Å continuum is used as a proxy for the ionizing extreme ultraviolet (EUV) continuum. Physical decoupling between the optical and ionizing continua (such as during the "reverberation holiday" or accretion disk wind events) can lead to apparent loss of information that is physical rather than methodological.
- **Interpolation and Seasonal Gaps:** Linear interpolation of the irregular campaign cadence onto a uniform 1-day grid acts as a low-pass filter and can preserve broad red-noise coherence across observational gaps. Although the isolated synthetic controls recover a 15-day input lag without appreciable mean bias, the multicomponent stress test shows large scatter and asymmetric lag errors.
- **Imperfect Term Isolation in Synthetic Tests:** The isolated single-driver, redundant-proxy, and nonlinear-joint controls recover their intended atoms, but the additive two-delay test does not: its intended unique terms are largest at the true lag in at most 1% of real-

izations. This limits how strongly we can interpret the detailed atom-by-atom structure of the real-data decompositions.

- **Finite-Sample Histogram Bias:** Mutual information estimators based on discrete histograms suffer from systematic sample-size bias (which tends to overestimate information and underestimate leaks). This is especially problematic in high-dimensional calculations (such as target-history conditioning) where joint 4D cells become sparsely populated.
- **Estimator Scope:** The KNN cross-check avoids joint histogram sparsity and supports the prompt-versus-long-lag significance conclusion, but it estimates total and conditional mutual information rather than the SURD atom partition. The detailed atom identities therefore remain specific to the histogram implementation.
- **Binning Sensitivity:** As documented in Section 4.11, changing the bin count and edge construction moves the conditional maximum from 37 to 102 d and strongly changes its amplitude. All tested settings remain globally non-significant, so only the null conclusion is stable.
- **Stationarity Assumptions:** The SURD analysis assumes that the coupling between the central engine and the BLR remains stationary over the 1744-day campaign. However, physical changes in the accretion rate, accretion disk wind structure, or BLR gas distribution can introduce non-stationarities that violate this assumption.

6 CONCLUSIONS

This paper presents an application of the Synergistic–Unique–Redundant Decomposition (SURD) algorithm to velocity-resolved AGN reverberation mapping data, focusing on the Hβ profile of NGC 5548. Our main conclusions are:

(i) The strict-overlap preprocessing pipeline limits edge effects and delivers a synchronized dataset over the JD offset 47512–49255 window.

(ii) In the exactly matched 999-realization circular-shift analysis, no lag survives empirical Benjamini–Hochberg correction and the four global maximum-statistic p-values are 0.873–1.000. The range-dependent SURD maxima therefore provide no significant evidence for physical long-lag delays.

(iii) The one-predictor continuum baseline recovers prompt peaks at 16 d for Blue and 15 d for Core. With 999 surrogates, their lag-scan global values are 0.009 and 0.001, and their three-target adjusted values are 0.0135 and 0.003. Red is not significant at 0.228.

(iv) The central detection grid uses 3400 realizations with separate 200 realization null calibration and evaluation samples. At $A = 0.6$, power is 0.945–0.970 and lag RMSE is 1.54–1.74 d, but process-timescale sensitivities preclude a universal threshold claim. The parameter A is a mixing coefficient, not a variance percentage.

(v) Every tested incremental directional scan is non-significant under both discrete and standardized KSG estimators. The blocked prediction sensitivity has only two usable strict-overlap blocks and gives negative incremental R^2 in both. The supported conclusion is nondetection under the declared aggregate null, not refutation of physical transport.

(vi) Native-cadence bidirectional ICCF gives FR/RSS centroid lags of approximately 19, 23, and 30 d for the blue, core, and red components, whereas the non-significant SURD maxima occur at 93–198 d. These statistics have different estimands and must not be interpreted as competing lag measurements.

(vii) Matched independent-red-noise ablations reproduce unstable normalized maxima with complete data and after applying the observing mask and interpolation. Histogram dimensionality, red-noise structure, sampling, interpolation, and normalization are jointly sufficient; the experiment does not identify one ingredient as the unique cause.

(viii) The minimum information leak for the line targets occurs at 1–9 days, much shorter than the SURD synergy maxima, but the leak minimum is not stable enough to treat as a precise lag measurement on its own. In the separate 1–120 d core-target conditioning test, the default conditioned maximum occurs at 73 d and is globally insignificant ($p_{\mathrm{global}} = 0.9802$). Across 3–8 equal-width and quantile bins, the maximum moves from 37 to 102 d, cell occupancy deteriorates, and every auxiliary global test remains non-significant ($p_{\mathrm{global}} = 0.64$–$1.00$).

(ix) Across 100 realizations, synthetic single-driver, redundant-proxy, and nonlinear-joint positive controls recover the input 15-day lag within ±5 d in every realization and identify the intended information atom in 99–100%. A harder additive two-delay stress test succeeds within ±5 d in only 67–71% and fails to isolate the intended unique atoms. The detailed real-data atom identities therefore remain exploratory.

(x) A continuous KNN estimator recovers globally significant prompt dependence for the three line targets but no significant 61–200 d maximum. KNN conditional mutual information is globally non-significant for $k = 3$, 5, and 10. The principal prompt-versus-long-lag conclusion therefore survives a non-histogram estimator, although the SURD atom identities do not receive an estimator-independent validation.

(xi) Native-cadence KNN scans of the independent Mrk 817 STORM 2 campaign show significant information at the published Lyα and C IV delays after a five-line correction, but unrestricted peak locations are generally sensitive to k and time-pairing tolerance. This independently confirms that information-curve maxima require prespecified tests and robustness controls before being interpreted as lags.

(xii) Although SURD provides a formally richer partition of predictive information than conventional lag estimators, this dataset contains no information atom that is both statistically significant and robust enough to establish additional astrophysical knowledge. Its demonstrated value in this study is methodological: it exposes the validation burden required before multivariate information structure is interpreted physically.

The present dataset does not support further exploratory searches for a positive SURD result without creating a substantial risk of post hoc selection. A credible test of added astrophysical value should instead use a predefined analysis on denser simultaneous optical, ultraviolet, X ray, and velocity resolved observations, ideally with a decomposition validated across independent estimators. The historical window analysed here lacks the coverage required for that test.

DATA AND CODE AVAILABILITY

The NGC 5548 velocity-resolved spectra and optical continuum data are available from the AGN Watch compilation (http://www.astronomy.ohio-state.edu/~agnwatch/). The Mrk 817 HST/COS light curves are available from the MAST AGN STORM 2 high-level science product (doi:10.17909/n734-k698). The analysis code and derived tables are available in the versioned repository at https://github.com/am610/AGN_SURD.

APPENDIX A: ARCHIVED JAVELIN DIAGNOSTIC

Previously generated JAVELIN chains are retained only as an archived exploratory diagnostic (Zu et al. 2011). Their configuration and convergence were not revalidated for this work, and they have no role in any quantitative result. The blue-wing chain is concentrated near 22.5 d with a smaller mode near 19.5 d; the core and red-wing chains are visibly multimodal. They therefore do not support unique single-lag summaries.

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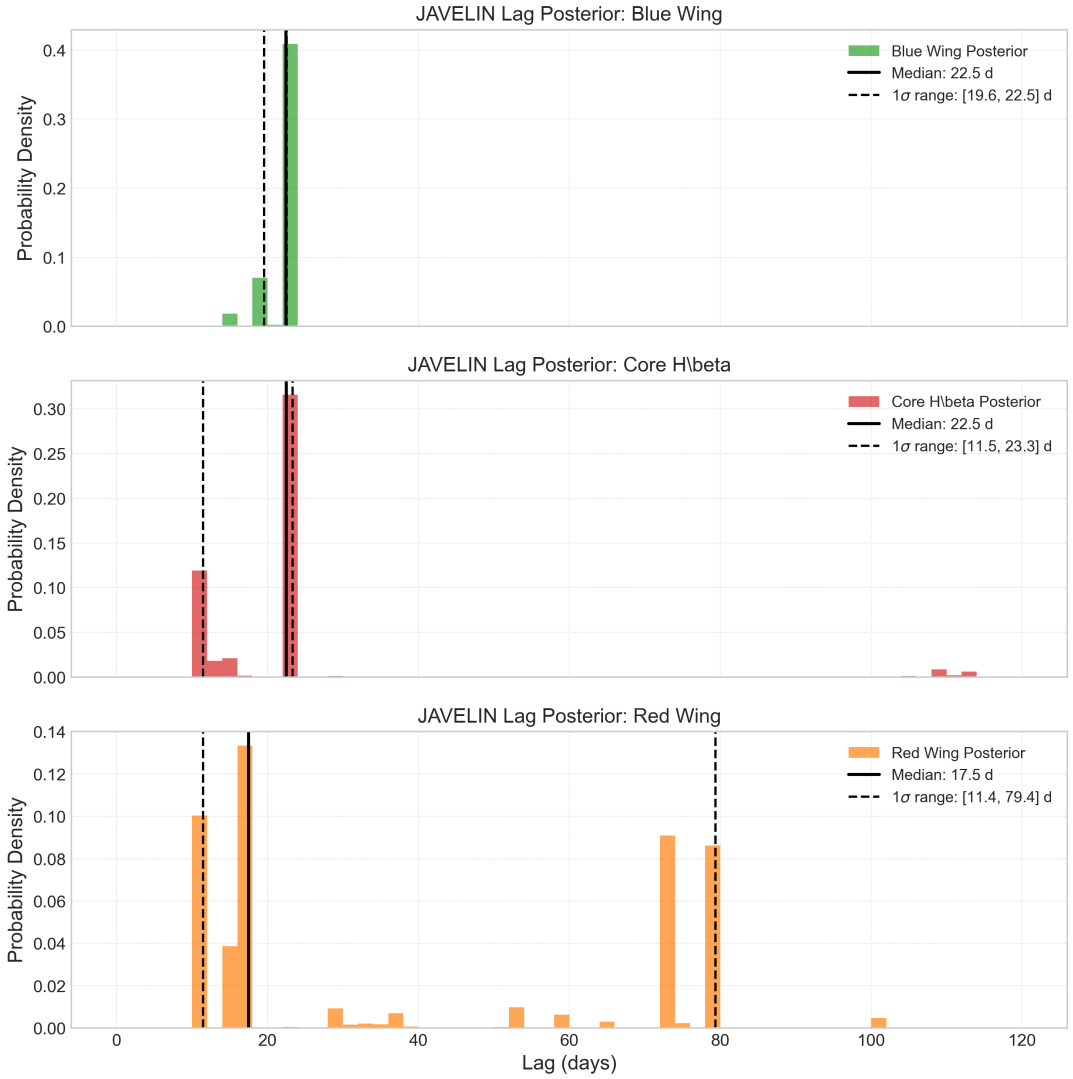


Figure A1. Archived JAVELIN lag-chain histograms. The multimodal chains are shown for transparency only and are not used for inference.